

/// Parameter Configuration Guide - North America

I-ETMS[®] Protect

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/// Revision History

Date	Revision	Description	Author
	0.1	This guide is based off the functionality in I-ETMS software build 6.5.1.1 corresponding to revision 2.1 of WCR-SYR-1097.	
12/1/2022	1.0	Incorporated the following DCRs: 174896 - Initial revision for I-ETMS Protect 172626 - port of 166246, 169642 169391 - Added TBC430-TBC436 165669 - added nav_module.cfg common file 177664 - added TBC437 and TBC438 177662 - added CFG40 and TBC427 177658 - added TBC428 and TBC429 176070 - updated ranges on TBC431-TBC436 176506 - support for metric speed parameters 162617 - Added CFB to railroads.cfg file 180985 - Added RUMO to railroads.cfg file 179801 - port of 153757	Wabtec
7/11/2023	1.1	Incorporated the following DCRs: 179794 (168115) - added SLRC under TRE in railroads.cfg file 179789 (171375) - added CFG42 and CFG43 180471 - added TBC442 and modified TBC260 178366 - modified ranges of TBC131, TBC133, TBC134, and TBC147 185529 - QoS settings for IVOC messages 171123 - added CFG45 183735 - modified range of TBC427 188085 - updated RUMO railroads.cfg entry 191279 - typo fix in TTL table 195781 - updated CFB railroads.cfg entry 175744 - Added TBC444 175609 - Added TBC445 204717 (203908) - modified ranges of TBC137, TBC140, TBC141, TBC297, TBC298, TBC308, TBC402	Wabtec
2/14/2024	1.2	Incorporated the following DCRs: 171125 - added TBC446 171124 - added TBC447 and CFG46 171127 - added CFG47 208502 (144593) - added TBC440 and TBC441, modified TBC225 175499 - added TBC448 and TBC449 205193 - added CFG49 205086 - updates to the valid range of TBC131, TBC133, TBC134, and TBC147 204532 - update to TBC429 for Guinea specific value 216504 - removed CFG47 205040 - added AML to railroads.cfg file	Wabtec
7/1/2025	1.3	Incorporated the following DCRs: 245472 - updates to TBC447 and CFG46 244766 - [Liberia ONLY] updated AML railroads.cfg entry 202208 - [Brazil ONLY] added MRS to railroads.cfg file 237249 - [Brazil ONLY] added VLI to railroads.cfg file	Wabtec

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		263098 (262675) - added TBC453,TBC454, TBC455 and CFG44	
8/20/2026	1.4	Incorporated the following DCRs: DCR 292238 (266960) - Added IANR reporting mark under CN SCAC 293350 (164853) - Format update 293776 (102897) - Added Narrative 293347 (165582) - Table 2.6 Update 293556 (157282) - Update TBC138 and added TBC439 293335 (166374) - Update TBC412 and CFG29 294381 (281397) - Added CFG50 292196 (285186) - Added MBTA SCAC 294225 (180022) - Added CFG48 DCR 264136 - Added CFG 66 DCR 291494 - Added CNR reporting mark DCR 387455 - Replaced incorrect Parameter CFG66 with CFG61 DCR 294499 (134069) - Modified TBC271 DCR 387974 (348451) - [Brazil ONLY] Updated MRS and RUMO railroads.cfg file DCR 321779 (196676) - Modified TBC412 DCR 264137 - Added TBC479 DCR 388494 (348450) - Updated TBC123 and TBC134 DCR 321761 (196073) - Added TBC 456 and CFG51 DCR 294855 (159177) - Added TBC450, removed TBC51, edited TBC40 DCR 294451 (102775) - Updated lower bound for parameters TBC140 and TBC141 DCR 392196 (390954) - Added NHN as reporting mark DCR 324205 (215483) - Updated units of TBC441 DCR 321807 (198939) - Added TBC457, TBC458, TBC459, TBC460, TBC461, TBC462 and CFG52 DCR 305109 (175236) - Revised configuration filesets DCR 324228 (219817) - Updated TBC163 DCR 388630 (387004) - Updated TBC260 and TBC442 DCR 309442 (195074) - Added TBC451 and TBC452 DCR 395411 (395408) [Brazil ONLY] - Updated TBC41 DCR 395699 [GUINEA ONLY]- Updated TBC168 DCR 396317 - Updates to the nav_module.cfg DCR 274854 - Addition of CFG65 DCR 386526 - Addition of CFG66, CFG67, CFG68 DCR 399547 - MRS Current speed decimal place display. Addition of CFG69 DCR 228749 - MRS Acceleration display. Addition of CFG70 DCR 294446 (262376) - Addition of CFG56 DCR 404929 - Adjustment of TBC288 in Guinea	Wabtec

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/// 1 Introduction

1.1 Purpose

This configuration guide provides configuration details for the On-board segment of the I-ETMS® system. Each section details the name of a configuration file and the type of information needed for that file.

1.2 Scope

This configuration guide contains a comprehensive list of configuration files required to operate the On-board segment. This configuration guide also documents the messaging name portion of the EMP address to be used when addressing messages to and from the various components hosted on the I-ETMS TMC.

1.3 References

The following documents are referenced by this user guide:

- I-ETMS® Protect PTC Office-Locomotive Segment Interface Control Document IV132.0, WEG-ICD-11097. Wabtec.
(See *I-ETMS Protect Compatibility and Compliance Matrix for applicable version*)
- ITC Time and Location ICD, Revision 1.2
- AAR Manual of Standards and Recommended Practices Section K-IV, AAR S-9356 Class D Messaging Rev 2010, Aug 2012.
- ITC Wayside-Locomotive ICD, revision 1.3, 10/16/2013
- AAR Manual of Standards and Recommended Practices, S-9354.V1.1 Edge Message Protocol
- IVOC-TMC-ICD, WEG-ICD-10723. Wabtec.
(See *I-ETMS Protect Compatibility and Compliance Matrix for applicable version*)

1.4 Definitions

Term	Description
Employing Railroad	The railroad that employs the person operating the I-ETMS on-board segment (i.e. the person who is logging in or has logged in)
Fileset	Collection of files that is transferred from an MDM Server to an MDM Client
MDM Server	Function of the office segment that communicates with MDM Clients.
MDM Client	Logical device to which filesets are downloaded (sent) and from which files are uploaded (received).
Operating Railroad	The railroad that dispatches the track at a particular location. This railroad is identified in the track data for the subdivision / district.
Owning Railroad	The railroad that owns or leases the locomotive on which the I-ETMS on-board segment is installed.

1.5 Acronyms

Acronym	Description
ACK	Positive Acknowledgement
ACP	Aspect Change Point or Asynchronous Communication Protocol
AG	Application Gateway
BCP	Brake Cylinder Pressure
BOS	Back Office Server
BPP	Brake Pipe Pressure
CDU	Cab Display Unit
CHM	Crash Hardened Memory
CHR	Crash Hardened Recorder
CI BOS	Crew Interaction BOS
CPU	Central Processing Unit

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Acronym	Description
DC	Direct Current
DIO	Discrete I/O Interface
DP	Distributed Power
EAB	Electronic Air Brake
EBI	Electronic Brake Interface
EM	Energy Management
EMP	Edge Message Protocol
EMT	Enter Main Track
EOT	End of Train
GPS	Global Positioning System
HP	Horsepower
ICD	Interface Control Document
ICE	Integrated Cab Electronics ®
ID	Identification
I-ETMS®	Interoperable – Electronic Train Management System
ILC	Integrated Locomotive Computer ®
IOC	Input / Output Concentrator
IP	Internet Protocol
ITC	Interoperable Train Control
KB	Kilobyte
KES	Key Exchange Server
LIG	Locomotive Interface Gateway
LSI	Locomotive Systems Integration
MB	Megabyte
MDA	Mobile Device Application
MDM	Mobile Device Manager
NAK	Negative Acknowledgement
PCS	Pneumatic Control Switch
PIN	Personal Identification Number
PPR	Pulses per Revolution
PSS	Pass Signal at Stop
PTC	Positive Train Control
QES	Q-Tron Excitation System
QoS	Quality of Service
RR	Railroad
RSM	Router Switch Module
SCAC	Standard Carrier Alpha Code
SG	Signal Group
TBC	To Be Configurable
THE	Train Handling Exception
TMC	Train Management Computer
TTL	Time to Live
VLAN	Virtual Local Area Network
WIU	Wayside Interface Unit
WSRS	Wayside Status Relay Service
XTE	Cross Track Error

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/// 2 Common Configuration Fileset

The common configuration filesets include:

- `params_COMMON.cfg`
- `<ABC>_ts_guide.cfg`
- `railroads.cfg`
- `ietms_qos.cfg`
- `em_safety_msgs.cfg`
- `train_restriction_types.cfg`
- `nav_module.cfg`

Changes to the parameters in any of these configuration filesets must first be approved by the interoperable railroads.

All common configurable files must be included in the fileset in order to operate the On-board segment. Should the on-board be unable to locate or verify a required .cfg file, the System Configuration check during initialization will not pass.

2.1 TBC Common Parameters

Configuration File: `params_COMMON.cfg`

The TBC common parameters contained in the `params_COMMON.cfg` file are configured for interoperability and used by all owning, employing, and operating railroads.

Note that for railroads operating in metric, all parameters representing speeds must be entered as kilometers per hour within the ranges defined in the table below for applicable parameters.

Note TBC common parameters may have separate valid ranges for imperial versus metric which are not numerically equivalent as the interoperability markets are geographically segregated.

NOTE Any change to these parameters must be approved by all interoperating railroads.

Table 2.1: TBC Common Parameters

TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC3	Authority Violation Timeout <i>Period for when an authority violation times out if an explicit report has not been received, after which crew will be prompted to verify whether the violation still exists.</i>	15	minutes	0-180
TBC12	Route Based Cab Restricted Speed Enforcement Delay Time <i>Utilized on route based cab signal track to calculate an enforcement delay equal to $(\text{current stopping time} \times ((\text{current speed} - \text{target speed}) / \text{current speed})) + \text{TBC12 seconds}$ when the cab signal changes and the train speed is faster than the speed associated with the new cab signal status.</i>	60	seconds	0-180
TBC14	Cab Signal Target Modification Time <i>Period of time since the calculated position uncertainty behind the leading edge of the train passed a facing signal, entered cab signal track, or exited a known cab signal dropout location after which the on-board segment will update its cab signal targets.</i>	13	seconds	0-180

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC26	Communication Flag Display Determination Factor <i>A multiplier to the fast poll rate that essentially determines the number of fast polls that may be missed before sending a poll registration request.</i>	2.1	n/a	0.1-4.0
TBC27	Conditional Authority Prompting Distance <i>The distance to the zero (0) mph conditional movement authority which the calculated position uncertainty ahead of the leading edge of the train must be within for the on-board to prompt crew to "Indicate Arrival of Engines"</i>	1500	feet	0-5000
TBC28	Crew Initiated Horn Protection Timeout <i>The on-board segment will not activate horn protection unless this minimum period of time has elapsed since the horn button was last pressed</i>	20	seconds	0-180
TBC29	Crossing Malfunction First Protection Prompt Distance <i>The distance to a malfunctioning grade crossing which the train must be within for the on-board to prompt crew</i>	3	miles	1-8
TBC30	Crossing Malfunction Warning Speed <i>The head end only speed limit set at a crossing when there is a malfunction of the crossing process.</i>	15	mph	1-15
		24	km/h	1-24
TBC40	Deliberation Distance <i>The distance in approach to a 0 mph signal target at which the on-board segment will</i> <i>prompt the crew to proceed past the signal displaying stop,</i> <i>use a pass signal at stop authority received from the BOS, or</i> <i>remove the target in the case of a stop and proceed signal indication</i> <i>assuming all other conditions for doing so are met.</i>	1500	feet	0-10000
TBC41	Depart Timeout <i>The time after a departure test has been performed before the next departure test will be required during initialization.</i>	96	hours	12-96
TBC55	Fast Poll Rate <i>The poll rate when determination distance is within TBC56 miles from the current onboard position.</i>	3	minutes	1-3
TBC56	Fast Poll Rate Determination Distance (from Locomotive) <i>If the shortest straight line distance between a locomotive and track in a subdivision is TBC56 or less, the on-board will assign a fast poll rate for that subdivision.</i>	50	miles	50-50
TBC57	Log File Archive Time <i>Time duration a log file is written to before being closed, archived, and replaced by a new log file.</i>	60	minutes	0-1000
TBC58	GPS Fix Validity Determination Time <i>The consecutive seconds that may elapse since the on-board segment last set the locomotive track location within a track segment based on a valid GPS fix before setting location and direction to unknown and generating appropriate targets.</i>	60	seconds	1-100
TBC78	Login Expiration <i>When the time duration 'since the last crew login' exceeds the login expiration value and the on-board state is CUT OUT or FAILED, the on-board system discards the stored employee ID, encrypted employee PIN, clearance number, and train ID.</i>	13	hours	12-100
TBC84	Maximum Empty Car Count	999	n/a	999-999

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC86	Maximum Equipment Speed Restriction	125	mph	125-125
		201	km/h	201-201
TBC87	Maximum Loaded Car Count	999	n/a	999-999
TBC89	Maximum Locomotive Count	24	n/a	24-24
TBC90	Maximum Locomotive Length	150	feet	150-350
TBC91	Maximum Locomotive Position	999	n/a	999-999
TBC92	Maximum Log Size <i>If a log file reaches the maximum log size, the log file is closed, archived, and replaced by a new log file.</i>	350	kb	1-1024
TBC93	Maximum Total Train Length	25000	feet	25000-25000
TBC95	Maximum Trailing Tonnage	45000	tons	45000-45000
TBC96	Medium Poll Rate <i>The poll rate when determination distance is between TBC56 and TBC97 from the current onboard position.</i>	10	minutes	3-20
TBC97	Medium Poll Rate Determination Distance (from Locomotive) <i>The upper limit of medium poll rate distance.</i>	100	miles	50-100
TBC98	Message Advance Limit <i>The acceptable limit for how far in the future the time stamp of message will be accepted.</i>	10	seconds	0-120
TBC99	Message Delay Limit <i>The acceptable limit for how far in the past the time stamp of message will be accepted.</i>	60	seconds	0-120
TBC100	Message Retry Count <i>The total number of times a message will be sent when the expected response is not received. Includes the initial message sent.</i>	3	n/a	0-3
TBC101	Message Timeout <i>The elapsed since the on-board segment sent a request message before it attempts a retry. Must be greater than the configured TTL for associated 2000 series messages.</i>	60	seconds	30-60
TBC103	Minimum Emergency Brake Reporting Speed <i>Minimum speed threshold value for emergency brake application report. Emergency brake application above this value generates a report.</i>	15	mph	0-125
		24	km/h	0-201
TBC104	Minimum Empty Car Count	0	cars	0-0
TBC106	Minimum Equipment Speed Restriction	0	mph	0-0
		0	km/h	0-0
TBC107	Minimum Loaded Car Count	0	cars	0-0
TBC109	Minimum Locomotive Count	1	locos	1-1
TBC110	Minimum Locomotive Length	40	feet	40-40
TBC111	Minimum Locomotive Position	1	n/a	1-1
TBC112	Minimum Position Logging Interval <i>The on-board system logs information from each GPS receiver at a minimum of this frequency.</i>	5	seconds	0-1000
TBC113	Minimum Total Train Length	40	feet	40-10000
TBC115	Minimum Trailing Tonnage	0	tons	0-0
TBC116	Moving Parking Flag Enforcement <i>When the parking function is enabled, if the locomotive moves for a duration equal to the moving parking flag enforcement value, the on-board system displays the braking banner, a 'Parking Brake On' brake prompt, and initiates a full service brake application.</i>	5	seconds	1-1000
TBC121	Poll Registration Interval	60	minutes	1-1000

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>The maximum elapsed time since the on-board segment sent a Poll Registration (02020) message for a subdivision/district before sending a new Poll Registration (02020) message..</i>			
TBC122	Position Report to Owning Railroad Frequency <i>Frequency the on-board system sends position report to owning railroad.</i>	15	minutes	5-60
TBC123	Position Report to Operating Railroads Frequency <i>Frequency the on-board system sends position report to operating railroad.</i>	5	minutes	0-1000
TBC124	Position Report Suppression Timeout <i>The minimum interval at which the on-board segment will send position reports.</i>	30	seconds	0-1000
TBC131	Predictive Removal Speed Warning Limit <i>If the on-board is predictively warning for a target, and the locomotive slows to within TBC131 mph over the speed of that target, the on-board will remove the warning.</i>	2	mph	0 to 125
		3	km/h	-8 to 8
TBC132	Freight Predictor Curve Nominal Warning Time <i>At current throttle and brake settings, the on-board will begin warning when it calculates there is TBC132 seconds before braking is required to be enforced. Changes in throttle and brake settings may cause the actual warning time to be longer or shorter than TBC132 seconds.</i>	45	seconds	0-180
TBC133	Overspeed Braking Limit <i>When predictively enforcing a target, the projected speed over the target speed at the target limit at which warning or braking may be initiated. Also, when predictively enforcing a target, the current speed over the target speed at which braking may be initiated.</i> <i>When reactively enforcing a target, the speed over the target speed at which braking is initiated.</i>	5	mph	0 to 5
		8	km/h	-8 to 8
TBC134	Overspeed Warning Limit <i>When predictively enforcing a target, the current speed over the target speed at which warning may be initiated.</i> <i>When reactively enforcing a target, the speed over the target speed at which warning is initiated.</i>	3	mph	0 to 5
		5	km/h	-8 to 8
TBC138	Minimum Distance to Target Next Signal More Permissive than SG-2 <i>When the leading edge of the train is in a restricted speed target or on non-PTC/PTC entry track, the on-board segment will generate a target for a facing signal with a status more permissive than a status in SG-2 when:</i> <i>The locomotive speed is less than or equal to target speed + TBC291 mph and the distance from the leading edge of the train to that signal is TBC138 feet OR</i> <i>The locomotive speed is greater than target speed + TBC291 mph and the distance from the leading edge of the train to that signal is the longer of:</i> <i>TBC138 feet or</i> <i>Stopping distance plus TBC439 feet</i>	3000	feet	0-10000

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC139	Seconds to Minutes Rounding Limit <i>Seconds to minutes rounding value for warning banner and enforcement braking. If greater than limit value, system rounds up to next higher minute. If less than limit value, system rounds down to next lower minute.</i>	30	seconds	1-59
TBC146	Slow Poll Rate <i>The poll rate to be used when the straight line distance to the that subdivisions track is greater than TBC97.</i>	60	minutes	20-60
TBC147	Soft Target Enforcement Activation Speed <i>The speed over the actual speed of the soft target that when the trains speed drops below this number, enforcement will become enabled for this target.</i>	3	mph	0 to 125
		5	km/h	-8 to 8
TBC148	Software Install Banner Timeout <i>During the installation of new software during initialization, the software install banner timeout is the time duration the "Installing new software, brakes will be applied" banner is displayed.</i>	10	seconds	0-1000
TBC156	Subdivision/District Non-Synchronization Determination Factor <i>A multiplier to the poll rate that essentially determines how long office synchronization may be lost.</i>	3.1	n/a	1.0-4.0
TBC157	Subdivision/District Non-Synchronization Target Enforcement Delay <i>When a non synchronized zero (0) speed target is generated, this is the delay time before it is enforce.</i>	30	seconds	0-1000
TBC161	Work Zone Target Receipt of Dataset / State Change Enforcement Delay <i>Enforcement delay ensures that enforcement braking does not occur for a 0 mph work zone target:</i> <i>until at least TBC161 seconds after the bulletin containing the work zone is received or</i> <i>until at least TBC161 seconds after the on-board segment state transition from DISENGAGED to ACTIVE.</i> <i>See also Work Zone Effective Time Enforcement Delat TBC249</i>	30	seconds	0-180
TBC162	Exit Notification Distance <i>The distance past the exit from PTC controlled track that the notification is displayed.</i>	1500	feet	0-10000
TBC163	The speed at which the train is considered stopped	0.1	mph	0.1-0.1
		0.2	km/h	0.2-0.2
TBC166	Second-to-Minute Conversion Threshold <i>When displaying the warning banner, the time at which the display changes from minutes to seconds.</i>	90	seconds	0-180
TBC167	Temporary Speed Restriction Receipt of Dataset Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC167 seconds after the on-board segment receives the bulletin or TBC248 after the temporary speed restriction goes into effect.</i>	30	seconds	0-180
TBC168	Train Handling Timeout <i>The elapsed time since the most recent train handling exception was detected, and the train handling exception prompt is still displayed</i>	30	seconds	0-1000
TBC171	Train Route Calculation Interval	60	seconds	1-1000

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>The interval at which the route is recalculated if nothing else has occurred to change or update the route.</i>			
TBC172	Train's Front Route Calculation Distance <i>Miles in front of train for which route is calculated.</i>	6	miles	5-8
TBC173	Train's Rear Route Calculation Distance <i>Miles to the rear of the train for which route is calculated.</i>	3	miles	2-8
TBC174	Unknown to Known Direction Determination Distance <i>The distance the train must travel when direction is unknown to determine direction of travel.</i>	30	feet	14-100
TBC175	WIUStatus Message Age Limit <i>The creation time stamp on WIU message must be less than this number of seconds old to be accepted by the onboard system.</i>	18	seconds	1-60
TBC177	Violation Report Interval <i>The interval at which violation reports are sent as long as the violation continues.</i>	5	minutes	0-1000
TBC178	Violation Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC178 seconds after the on-board segment receives the violation.</i>	75	seconds	0-180
TBC182	Prompt Display Distance <i>Distance from a joint authority or Work Zone target within which the on-board will prompt.</i>	3	miles	2-8
TBC183	Target Display Time <i>The amount of time in advance of the start time of a bulletin of any type other than work zone that the on-board segment will generate a target for that bulletin.</i>	60	seconds	0-180
TBC184	Joint Movement Authority Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a 0 mph joint movement authority target</i> <i>until at least TBC184 seconds after the movement authority is received if the leading edge of the train is inside the joint limits or</i> <i>until at least TBC184 seconds after the on-board segment state transitions from DISENGAGED to ACTIVE.</i>	30	seconds	0-180
TBC185	XTE Limit <i>Cross Track Error Limit</i>	17	meters	5-17
TBC186	Maximum Axle Count	3996	n/a	3996-3996
TBC187	Maximum Locomotive Horsepower	10000	hp	10000-10000
TBC188	Maximum Locomotive Tonnage	400	tons	400-400
TBC189	Maximum Operative Brake Count	999	brakes	999-999
TBC190	Minimum Axle Count	0	axles	0-0
TBC191	Minimum Locomotive Horsepower	0	hp	0-0
TBC192	Minimum Locomotive Tonnage	50	tons	50-50
TBC193	Minimum Operative Brake Count	0	brakes	0-0
TBC194	BeaconOn Interval <i>The minimum interval at which the on-board segment will send Beacon Request messages to WIU.</i>	19	seconds	1-1000
TBC195	BeaconOn Stagger Interval <i>In order to avoid multiple locomotives sending BeaconOn messages to a WIU and potentially disrupting communications, the BeaconOn Stagger Interval is in place to allow the closest locomotive to send the message. Upon</i>	1	sec/mile	1-100

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>receipt of a WIUStatus message indicating that the WIU is about to stop beaconing, the on-board segment will wait for a period equaling: Route distance between device and leading edge of the train in miles x TBC195 seconds/mile before sending a Beacon Request message.</i>			
TBC196	GetWIUStatus Duration <i>If the known status of the WIU is within TBC196 seconds of expiring and the train is within stopping distance of the device and the on-board segment is not braking for a target generated for that device, the on-board will send the GetWIUStatus message.</i>	4	seconds	0-1000
TBC197	GetWIUStatus Interval <i>Time to prevent too frequent WIU status.</i>	1	seconds	0-1000
TBC199	Minimum Copyright Screen Timeout <i>The minimum time the Copyright screen should be displayed.</i>	30	seconds	0-1000
TBC201	Maximum Along Track Uncertainty Limit	50	meters	1-57
TBC202	Maximum Locomotive Weight	800000	pounds	800000-800000
TBC203	Minimum Locomotive Weight	100000	pounds	100000-100000
TBC204	Maximum Wheel Diameter	46.00	inches	46.00-46.00
TBC205	Minimum Wheel Diameter	6.00	inches	6.00-6.00
TBC206	Maximum Wheel Tachometer PPR	1200	ppr	1200-1200
TBC207	Minimum Wheel Tachometer PPR	4	ppr	4-4
TBC208	Minimum Penalty Brake Set Time <i>During a brake test, the penalty brake must be activated for at least TBC208 seconds before the Toggle Emergency key can be utilized.</i>	10	seconds	1-1000
TBC212	Poll Rate Assignment Timeout <i>Poll rate will be assigned once every TBC212 minutes.</i>	3	minutes	1-16
TBC213	Fault Report Elapse Time <i>If the on-board has not sent a Locomotive Fault Report since a fault status change and the time since the fault status change is at least TBC213 seconds, the on-board will send a Fault Report</i>	10	seconds	1-1000
TBC214	Enter Intermittent Fault State Time <i>When the status of a fault transitions between clear and active more than once within TBC214 minutes, the on-board segment declares the fault as intermittent.</i>	1	minutes	1-60
TBC215	Exit Intermittent Fault State Time <i>When an intermittent fault has been cleared for at least TBC215 minutes, the on-board segment declares the intermittent fault as cleared.</i>	15	minutes	1-60
TBC220	Minimum Total Braking Force	0	pounds	0-0
TBC221	Maximum Total Braking Force	30,000,000	pounds	30,000,000-30,000,000
TBC223	Maximum PCS Delay <i>If TBC223 seconds elapse during the departure test and the PCS has not activated, the brake test will fail.</i>	10	seconds	1-20
TBC224	Elapse time for sending Train Control Status Data (0500) message <i>On-board will send a PTC Status Data message once every TBC224 seconds to Energy Management</i>	5	seconds	5-5

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC225	Elapse time for receiving Energy Management Status (0600) message. Also used for determining EM Communication Loss <i>If the on-board does not receive an Energy Management Status message within TBC225 seconds of connecting to Energy Management, the on-board will reset the Class D connection and. declare an EM Communication Loss.</i>	15	seconds	15-15
TBC227	Periodic time for sending Current Position (0530) message <i>On-board will send a Current Position message once every TBC227 seconds to Energy Management.</i>	5	seconds	5-5
TBC228	Distance for sending Current Position (0530) message <i>On-board will send a Current Position message to Energy Management if the train has moved TBC228 feet since the previous Current Position message.</i>	500	feet	500-500
TBC229	Periodic time for sending Train Route (0531) message <i>If on-board has track location and direction, it will send a Train Route message once every TBC229 seconds to Energy Management</i>	15	seconds	15-15
TBC230	Periodic time for sending Target List (0532) message <i>If on-board has track location and direction, it will send a Target List message once every TBC230 seconds to Energy Management</i>	15	seconds	15-15
TBC231	Elapse time for receiving EM Display Information <i>If on-board does not receive an Energy Management advisory message for TBC231 seconds, the on-board will remove the applicable feature from the display</i>	5	seconds	5-5
TBC232	Elapse time for receiving Crew and Train Data Acknowledge (0624) message <i>If TBC232 seconds has elapsed since on-board sent a Crew and Train Data message without receiving an acknowledgement, the on-board will resend the message.</i>	1	seconds	1-1
TBC233	Elapse time for receiving Target List Acknowledge (0632) message <i>If TBC233 seconds has elapsed since on-board sent a Target List message without receiving an acknowledgement, the on-board will resend the message.</i>	1	seconds	1-1
TBC234	Minimum time elapsed prior to emergency brake application <i>The time a predictive full-service brake application must be activated before the on-board initiates an emergency brake application</i>	10	seconds	0-10
TBC235	Minimum speed for emergency brake application <i>The train speed must be above TBC235 for the on-board to transition from predictive full-service brake application to emergency brake application.</i>	2	mph	0-2
		3	km/h	0-3
TBC236	Emergency reserve threshold <i>During the departure test, the front BPP must remain above TBC236 psi in order for the on-board segment to detect that emergency brakes will be available for any subsequent enforcement braking.</i>	40	psi	35-45
TBC237	Software installation timeout <i>The maximum time allowed for a successful software installation.</i>	5	minutes	5-5
TBC238	Fast Owning Road MDM Fileset Synchronization Rate <i>The rate at which the on-board segment will contact the</i>	15	minutes	15-15

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>MDM Server of the owning/leasing railroad when the on-board segment is in a controlling state.</i>			
TBC239	Medium Owning Road MDM Fileset Synchronization Rate <i>The rate at which the on-board segment will contact the MDM Server of the owning/leasing railroad when the on-board segment is not in a controlling state.</i>	60	minutes	60-60
TBC240	Operating and Employing Road MDM Fileset Synchronization Rate <i>The rate at which the on-board segment will contact the MDM Server of the operating or employing railroad.</i>	15	minutes	15-15
TBC241	Slow MDM Fileset Synchronization Rate <i>The rate at which the on-board segment will contact the MDM Server of all other railroads contained in the common configuration fileset currently loaded.</i>	24	hours	24-24
TBC242	Track Out of Service Receipt of Dataset Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC242 seconds after the on-board segment receives the bulletin or TBC250 seconds after the bulletin goes into effect.</i>	30	seconds	0-180
TBC243	Grade Crossing Activation Failure Receipt of Dataset Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC243 seconds after the bulletin containing the grade crossing warning system malfunction is received or TBC251 seconds after the bulletin containing the grade crossing warning system malfunction goes into effect.</i>	30	seconds	0-180
TBC244	Grade Crossing False Activation Receipt of Dataset / State Change Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a grade crossing false activation target:</i> <i>until at least TBC244 seconds after the bulletin containing the grade crossing warning system malfunction is received or</i> <i>until at least TBC244 seconds after the on-board segment state transition from DISENGAGED to ACTIVE.</i> <i>See also Grade Crossing False Activation Effective Time Enforcement Delay TBC252.</i>	30	seconds	0-180
TBC245	Maximum Total Car Count	999	n/a	999-999
TBC246	Departure Test Start Minimum Threshold without Feed Valve Setting <i>Front brake pipe pressure must be above this number to initiate a brake test.</i>	80	psi	60-110
TBC247	Software Download Time Limit <i>Time at which the software download clock rolls over back to 0 Minutes 0 seconds.</i>	99	minutes	0-99
TBC248	Temporary Speed Restriction Effective Time Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC167 seconds after the on-board segment receives the bulletin or TBC248 after the temporary speed restriction goes into effect.</i>	30	seconds	0-180
TBC249	Work Zone Effective Time Enforcement Delay	30	seconds	0-180

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>The enforcement delay ensures that enforcement braking does not occur for a 0 mph work zone target until at least TBC249 seconds after the work zone goes into effect. See also Work Zone Target Receipt of Dataset / State Change Enforcement Delay TBC161.</i>			
TBC250	Track Out of Service Effective Time Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC242 seconds after the on-board segment receives the bulletin or TBC250 seconds after the bulletin goes into effect.</i>	30	seconds	0-180
TBC251	Grade Crossing Activation Failure Effective Time Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC243 seconds after the bulletin containing the grade crossing warning system malfunction is received or TBC251 seconds after the bulletin containing the grade crossing warning system malfunction goes into effect.</i>	30	seconds	0-180
TBC252	Grade Crossing False Activation Effective Time Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a grade crossing false activation target until at least TBC252 seconds after the bulletin containing the grade crossing warning system malfunction goes into effect. See also Grade Crossing False Activation Receipt of Dataset / State Change Enforcement Delay TBC244.</i>	30	seconds	0-180
TBC253	In-Train Brake Cylinder Pressures <i>The upper limit of display of in train brake cylinder pressure. Any BCP above TBC253 will be displayed at TBC253.</i>	50	psi	40-60
TBC254	Duplicate BOS Message Detection Period <i>On-board discards any message with same type, number, address, and integrity as a previously accepted message received within TBC254 seconds. When sending messages to BOS, on-board will generate a different correlation ID from any ID generated in the last TBC254 seconds.</i>	60	seconds	0-1000
TBC255	Violation No Longer Reported Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC255 seconds after the violation is no longer reported.</i>	75	seconds	0-180
TBC256	WIUStatus Message Advance Limit <i>On-board will discard any WIUStatus message with a creation time more than TBC256 seconds in the future of the time the message was received.</i>	4	seconds	0-60
TBC257	Passenger Predictor Curve Nominal Warning Time <i>At current throttle and brake settings, the on-board will begin warning when it calculates there is TBC257 seconds before braking is required to be enforced.</i>	30	seconds	0-180
TBC258	Commuter Predictor Curve Nominal Warning Time <i>At current throttle and brake settings, the on-board will begin warning when it calculates there is TBC258 seconds before braking is required to be enforced.</i>	30	seconds	0-180
TBC259	Maximum Log File Retention <i>Maximum amount of time the on-board keeps log files</i>	30	days	1-30

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
TBC260	Maximum CDU Log File Storage Capacity <i>On-board deletes oldest files when stored CDU logs exceed TBC260 MB</i>	5	MB	1-30
TBC261	WSRS Subscription Timeframe <i>On-board will remain subscribed to a WSRS for TBC261 seconds from the time of sending a subscribe message. Any attempts to unsubscribe will continue for this duration.</i>	900	seconds	60-3600
TBC262	Elapsed time since receiving WSRS Subscription Confirmation message <i>If a device monitored by a WIU via WSRS is still on the train's calculated route after TBC262 seconds, the on-board will send another WSRS Subscribe message. Note: this time frame should be less than WSRS Subscription Timeframe set in message.</i>	810	seconds	60-3500
TBC263	WSRS Subscription Message Timeout <i>On-board will resend the WSRS Subscribe message if no confirmation is received within TBC263 seconds</i>	60	seconds	0-60
TBC264	WSRS Unsubscribe Message Timeout <i>On-board will resend the WSRS Unsubscribe message if no confirmation is received within TBC264 seconds, and less than TBC261 seconds has elapsed since a WSRS Subscribe message has been sent</i>	60	seconds	0-60
TBC266	WSRS Backup Enabled subscription timeout due to loss of WIUStatus (5100) messages <i>On-board will send a WSRS Subscribe message to a WSRS backup enabled WIU if it has not received a valid WIUStatus message within the past TBC266 seconds</i>	10	seconds	0-16
TBC269	Approach Lighting Timeout <i>Minimum time since passing facing signal in approach to a dark, absolute, intermediate stop and proceed, or intermediate restricted signal before target generation</i>	30	seconds	1-30
TBC270	Crossing Malfunction Activation Failure Second Protection Prompt Distance <i>After the crew has indicated once that a crossing in a crossing activation failure target is not protected, the distance to that target the train must be within for the on-board segment to display a second prompt to indicate that the crossing is protected.</i>	1000	feet	0-10,000
TBC271	Allowable Rollback Distance <i>Distance train can travel in opposite direction before on-board sets location and direction to unknown.</i>	50	feet	0-1000
TBC274	Unknown Location Target Enforcement Delay	30	seconds	0-1000
TBC276	Position Uncertainty Target Enforcement Delay	30	seconds	0-1000
TBC278	Unknown Direction of Movement Target Enforcement Delay	30	seconds	0-1000
TBC280	Subdivision/District Non-Synchronization Target Prompting Distance	15,840	feet	0-20,000
TBC287	Departure Test Start Minimum Threshold with Feed Valve Setting <i>Front brake pipe pressure must be above the feed valve setting minus this number to initiate a brake test.</i>	15	psi	0-40
TBC288	Switch Resolution Prompt Delay <i>On-board will wait TBC288 seconds after the position of a monitored switch becomes unknown before prompting. This prevents a switch resolution prompt from being</i>	30	seconds	0-40

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>displayed, for example, when a hand throw switch is in the process of being thrown.</i>			
TBC291	Overspeed Warning Limit for Restricted Speed Targets <i>When predictively enforcing a restricted speed target, the current speed in relation to the target speed at which warning may be initiated.</i> <i>When reactively enforcing a restricted speed target, the speed in relation to the target speed at which warning is initiated.</i>	0	mph	-5 to 5
		0	km/h	-8 to 8
TBC292	Overspeed Braking Limit for Restricted Speed Targets <i>When predictively enforcing a restricted speed target, the projected speed in relation to target speed at the target limit at which warning or braking may be initiated. Also, when predictively enforcing a restricted speed target, the current speed in relation to the target speed at which braking may be initiated.</i> <i>When reactively enforcing a restricted speed target, the speed in relation to the target speed at which braking is initiated.</i>	2	mph	-5 to 5
		3	km/h	-8 to 8
TBC293	Predictive Warning Removal Limit for Restricted Speed Targets <i>The current speed in relation to the target speed at which a warning is removed.</i>	-1	mph	-5 to 5
		-2	km/h	-8 to 8
TBC294	Speed Enforcement Cab Signal Freight Speed Enforcement Delay Time <i>Utilized on speed enforcement cab signal track to calculate an enforcement delay for a freight train equal to:</i> <i>(current stopping time x ((current speed – target speed) / current speed)) + TBC294 seconds</i> <i>when the cab signal changes and the train speed is faster than the speed associated with the new cab signal status.</i>	72	seconds	0-180
TBC295	Speed Enforcement Cab Signal Passenger Speed Enforcement Delay Time <i>Utilized on speed enforcement cab signal track to calculate an enforcement delay for a passenger train equal to:</i> <i>(current stopping time x ((current speed – target speed) / current speed)) + TBC295 seconds</i> <i>when the cab signal changes and the train speed is faster than the speed associated with the new cab signal status.</i>	72	seconds	0-180
TBC296	Cab Signal Restricted Distance <i>The train will be required to travel the greater of the train length or this many feet after a restricting cab signal indication upgrades.</i>	500	feet	0-1500
TBC297	Clear to Next Interlocking Speed for Freight <i>The maximum authorized speed when a clear to next interlocking indication is received from a signal.</i>	79	mph	1-125
		127	km/h	1-201
TBC298	Clear to Next Interlocking Speed for Passenger <i>The maximum authorized speed when a clear to next interlocking indication is received from a signal.</i>	79	mph	1-125
		127	km/h	1-201
TBC299	MDM Maximum Data Transmission Size	1500	bytes	0-2048
TBC300	MDM Transaction Timeout	300	seconds	0-1000
TBC301	Maximum Speed for Time Based Horn Protection <i>The speed at or below which the on-board segment logic for horn protection is based on time from crossing and above which the on-board segment logic for horn</i>	45	mph	0-125
		72	km/h	0-201

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>protection is based on time and distance from crossing</i>			
TBC302	Horn Protection Activation Distance Threshold <i>The distance from a crossing that the on-board segment will activate horn protection when the train speed is above Maximum Speed for Time Based Horn Protection</i>	1320	feet	0-5280
TBC303	Unknown Switch Position Target Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for an unknown switch position target</i> <i>Until at least TBC303 seconds after the on-board segment state transitions from DISENGAGED to ACTIVE and</i> <i>Until at least TBC303 seconds after the unknown switch position target is generated inside a restricted speed cab signal target.</i>	30	seconds	0-180
TBC304	Conditional Movement Authority Target Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a conditional movement authority target until at least TBC304 seconds after the on-board segment state transitions from DISENGAGED to ACTIVE.</i>	30	seconds	0-180
TBC305	Restricted Speed Signal Target Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a restricted speed signal target until at least (current stopping time x ((current speed - target speed) / current speed)) + TBC305 seconds after the on-board segment state transitions from DISENGAGED to ACTIVE.</i>	60	seconds	0-180
TBC306	CHM Class D Data Ack Timeout <i>The number of milliseconds to wait for a Data Acknowledgement before timing out</i>	30000	ms	0-60000
TBC309	Unintended State Change Speed Reduction Period <i>The period for which the on-board segment will warn before initiating a full-service brake application if the train speed is above TBC308/TBC402 mph following an "unintended" transition from the ACTIVE state to the DISENGAGED, CUT OUT or FAILED state.</i>	120	seconds	0-180
TBC310	Tolerance for Bunching Moves <i>The distance that the on-board segment will allow the train to move opposite the direction authorized by a directional form-based authority without enforcement braking when CFG1 is disabled.</i> <i>Also the distance the on-board segment will allow the train to move after changing direction in a work zone without enforcement braking when CFG3 is disabled.</i> <i>Also the distance the on-board segment will allow the train to move opposite the direction in which it entered a signal block before enforcing restricted speed until the leading edge of the train leaves that signal block, regardless of any subsequent change in direction.</i>	50	feet	0-100
TBC312	PTC Suspension Receipt of Dataset / State Change Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC312 seconds after the on-board segment receives the bulletin or TBC313 after the</i>	30	seconds	0-180

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>PTC suspension bulletin goes into effect.</i>			
TBC313	PTC Suspension Effective Time Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur until at least TBC312 seconds after the on-board segment receives the bulletin or TBC313 after the PTC suspension bulletin goes into effect.</i>	30	seconds	0-180
TBC314	PTC Suspension Target Prompting Distance	3	miles	2-8
TBC380	Target Approach Management Freight Approach Distance <i>The distance from a zero speed target at which the on-board segment may suspend predictive enforcement for a zero speed target per the target approach management function on a train of type "freight" or "intermodal".</i>	1000	feet	1-1000
TBC381	Target Approach Management Freight Maximum Speed <i>The speed at or below which the on-board segment may suspend predictive enforcement for a zero speed target per the target approach management function on a train of type "freight" or "intermodal".</i>	10	mph	0-10
		16	km/h	1-16
TBC383	Reverse Movement outside Signal Block Prompt Distance <i>The distance to the signal governing exit from a signal block the train must be within for the on-board segment to display a prompt to indicate that they have permission to make such a move.</i>	2000	feet	0-5000
TBC384	Maximum distance for retaining recorded locomotive track location <i>The maximum distance the on-board segment will allow the train to move in the Initializing state before discarding any locomotive track location it has retained in memory. Once discarded that locomotive track location is no longer available when the on-board segment determines the locomotive track location upon transition out of the Initializing state.</i>	5	meters	1-5
TBC385	Signal Target Removal Distance When Cab Signal More Permissive <i>The distance the locomotive must travel past a restricted speed signal target, a restricted speed enter main track target, or a SG-3 soft target before a more permissive cab signal target will be considered to take precedence by the on-board.</i>	1000	feet	0-10,000
TBC386	Reactive Braking Hold Off for Non-zero Speed Targets	3	seconds	0-3
TBC387	Activate "Soft" Target Period	3	seconds	1-3
TBC388	Minimum Nominal Warning Time <i>The on-board segment adjusts its warning distance calculation to ensure TBC388 seconds of nominal warning time prior to predictive enforcement braking.</i>	30	seconds	0-180
TBC389	Work Zone Minimum Reactive Warning	10	seconds	0-30
TBC390	Aggregate CRC Mismatch and Composite CRC Mismatch Hold Off Time <i>Time allowed for an aggregate CRC mismatch or a composite CRC mismatch to be cleared before the on-board segment sets the subdivision/district to "non synchronized."</i> <i>A value of 0 indicates that there is no hold off time and the on-board segment sets the subdivision/district to "non</i>	9	seconds	0-60

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>synchronized" immediately upon detection of an aggregate CRC mismatch or a composite CRC mismatch. Note that the on-board segment will detect a composite CRC mismatch only when CFG37 is enabled for the railroad operating the subdivision/district.</i>			
TBC391	Approaching Exit Notification Distance <i>The distance in approach to the exit from PTC controlled track that the notification is displayed.</i>	10000	feet	0-10000
TBC392	Tolerance for Train Handling in Work Zone <i>The time that the on-board segment will allow the train to stop within a work zone the crew has already confirmed permission to enter before again generating a zero speed target over that work zone.</i>	10	seconds	1-30
TBC393	Target Approach Management Passenger Approach Distance <i>The distance from a zero speed target at which the on-board segment may suspend predictive enforcement for a zero speed target per the target approach management function on a train of type "passenger", "commuter", "high speed passenger", or "tilt train".</i>	1000	feet	1-1000
TBC394	Target Approach Management Passenger Maximum Speed <i>The speed at or below which the on-board segment may suspend predictive enforcement for a zero speed target per the target approach management function on a train of type "passenger", "commuter", "high speed passenger", or "tilt train".</i>	10	mph	0-10
		16	km/h	0-16
TBC395	Expanded XTE Limit <i>A larger cross track limit used to increase the number of potential track options when the crew is selecting track location and no track options are within TBC185 of the locomotive's GPS position.</i>	25	meters	18-30
TBC396	Acceptable Along-Route Distance for GPS Fix <i>The along-route distance a valid GPS fix can be from the current locomotive track location before setting the location to unknown in DISENGAGED or generating an unknown location target in ACTIVE.</i>	200	feet	0-10,000
TBC397	Prompt for continued movement in Restricted state distance <i>The distance that a locomotive may travel since the locomotive last stopped or the crew last acknowledged a prompt for continued movement in Restricted state before being prompted to continue movement in Restricted state.</i>	26,400	feet	100-52,800
TBC398	Prompt response time for continued movement in Restricted state <i>The time allowed for the crew to respond to a prompt for continued movement in Restricted state before the on-board segment will command a full-service brake application.</i>	30	seconds	0-60
TBC399	Maximum leading end track solution count	5	N/A	1-10
TBC400	Transition to Restricted state enforcement delay	120	seconds	0-180
TBC401	Pending Prompts Timeout <i>Maximum time "Select prompt to answer" prompt and "Pending Prompts" list can be displayed</i>	60	seconds	30-180
TBC403	Unintended State Change Prompt Response Period	120	seconds	0-180

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>The period for which the on-board segment will prompt the crew to acknowledge that reduced speed is required if the train speed is at or below TBC308/TBC402 mph following an "unintended" transition from the ACTIVE state to the DISENGAGED, CUT OUT or FAILED state.</i>			
TBC411	Processing Allowance for Accepting Wayside Signal Status while Moving <i>The on-board segment adds a buffer of TBC411 seconds at the train's current speed to the distance in advance of the leading edge of the train where the on-board segment will accept wayside signal status. This allows processing of the signal status to be completed by the time that the signal is within the calculated position uncertainty of the leading edge of the train.</i>	1	seconds	1-5
TBC412	Hold Off for Signals within the Calculated Position Uncertainty of the Leading Edge of the Train When Stopped <i>When the locomotive has been stopped for at least TBC412 seconds, the on-board segment will accept status updates from wayside signals within the calculated position uncertainty of the leading edge of the train and update its signal targets accordingly.</i> <i>If the locomotive has been stopped for at least TBC412 seconds and the on-board segment state is Disengaged, the on-board segment will not:</i> <i>Replace a zero (0) mph signal based target with a restricted speed signal based target when the leading edge of the train is within the calculated position uncertainty of that target</i> <i>Generate a becoming located signal target for a signal block within the calculated position uncertainty in advance of the leading edge of the train</i> <i>The locomotive must be stopped for at least TBC412 seconds for the on-board segment to store the last known signal target status upon a transition to Cut Out or Restricted state.</i> <i>If the locomotive has been stopped for at least TBC412 seconds and the on-board segment state is Active and the leading edge of the train is within calculated position uncertainty in approach to or inside of a zero (0) mph signal target or zero (0) mph unknown signal target and prompting is not enabled in the track data for the signal associated with that target, the on-board segment will prompt the crew that the signal status is preventing train movement.</i>	3	seconds	1-3
TBC413	Route based cab signal dropout target generation delay <i>The length of time that the route based cab signal status must be SG-2 before the on-board segment will update its route based cab signal targets to allow for short cab signal dropouts outside of a defined cab signal dropout location.</i>	5	seconds	0-180
TBC414	Speed enforcement cab signal dropout target generation delay <i>The length of time that the speed enforcement cab signal code rate must be zero (0) before the on-board segment will update its speed enforcement cab signal targets to</i>	5	seconds	0-180

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>allow for short cab signal dropouts outside of a defined cab signal dropout location.</i>			
TBC422	Speed increase at which the BPP Setpoint may be reset	10	mph	1-10
		16	km/h	1-16
TBC423	Braking Calculation Failure Target Enforcement Delay	30	seconds	1 - 1000
TBC428	Track Solution Update Interval <i>The amount of time before the on-board segment will check for a new track solution when the on-board segment is displaying a possible track solution.</i>	5	seconds	1-10
TBC429	Track Location Solution Uncertainty Limit <i>The maximum allowed size of the Track Location Solution Uncertainty. This limit is used to avoid automatic track selection when the train is occupying unmapped track or there is a sophisticated spoofing attack.</i>	90 90 (Brazil) 180 (Guinea)	inches	78-180
TBC437	Heading Uncertainty Limit <i>The maximum arc measurement of a calculated heading uncertainty arc.</i>	180	degrees	90-180
TBC438	Matching Heading Limit <i>The maximum difference in heading on either side of the track data heading that is considered a match.</i>	60	degrees	1-90
TBC439	Buffer in Addition to Stopping Distance to Target Next Signal More Permissive than SG-2 <i>When the leading edge of the train is in a restricted speed target or on non-PTC/PTC entry track, the on-board segment will generate a target for a facing signal with a status more permissive than a status in SG-2 when the locomotive speed is greater than target speed + TBC291 mph and the distance from the leading edge of the train to that signal is the longer of:</i> • TBC138 feet or • Stopping distance plus TBC439 feet	1000	feet	0-10000
TBC442	Maximum CPU-1900 Log File Storage Capacity <i>On-board deletes oldest files when stored CPU-1900 logs exceed TBC442 MB</i>	30	MB	1-256
TBC449	Train's Rear Route Calculation Distance from Rear-End Track Location <i>Distance from the rear-end track location which is used for route calculation.</i>	1500	feet	1-1500
TBC450	Enter Main Track Prompting Distance <i>The distance in approach to an enter main track target or at which the on-board segment will prompt the crew to indicate the train has authority to occupy controlled track or that the train did not clear controlled track.</i>	3000	feet	0-4000
TBC451	Consist Related Speed Enforcement Delay <i>The enforcement delay ensures that enforcement braking does not occur for a consist related speed target (equipment speed restriction, track based speed restriction, or train restriction) until at least TBC451 seconds after the target is generated following an update to the summary train consist.</i>	30	seconds	0-180
TBC452	Clear enforcement delay period <i>The on-board segment clears any remaining enforcement delay for a zero (0) mph target when the locomotive has</i>	3	seconds	1-3

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TBC#	Description	Common Value North America	Units	Valid Range (min-max)
	<i>been stopped for TBC452 seconds. The on-board segment clears any remaining enforcement delay for a non-zero mph target when the locomotive speed has been below the target speed plus the overspeed warning limit (TBC134 for non-restricted speed targets, TBC291 for restricted speed targets) for a continuous period of TBC452 seconds.</i>			
TBC453	PTC Status and Energy Management Status Logging Interval <i>The interval at which the PTC Status (0500) message is logged when the contents of the message are not changing. The interval at which the Energy Management Status (0600) message is logged when the contents of the message are not changing.</i>	30	seconds	0-180
TBC454	Energy Management In-Train Forces Logging Interval <i>The interval at which the In-Train Forces (0625) message is logged.</i>	30	seconds	0-180
TBC455	Energy Management Current Position Logging Interval <i>The interval at which the Current Position (0530) message is logged.</i>	30	seconds	0-180
TBC457	Percent of Operative Brakes per Four Axles	1	percent	1-100
TBC458	Minimum Empty Car Tonnage	1	tons	1-1
TBC459	Minimum Number of Axles per Car	2	N/A	2-2
TBC460	Minimum Length of Car	1	feet	1-1
TBC461	Maximum Number of Axles per Locomotive	8	N/A	8-8
TBC462	Minimum Number of Axles per Locomotive	4	N/A	4-4

2.2 Troubleshooting Guides

Configuration File: <ABC>_ts_guide.cfg

NOTE The <ABC> characters in the *actual* configuration files are replaced with the name of the corresponding diagnostics page.

The <ABC>_ts_guide.cfg files provide the help text for the on-board I-ETMS diagnostics troubleshooting guide pages. Any change to the text in these files must be approved by Wabtec and the interoperable railroads.

Table 2.2: Troubleshooting Guides

Troubleshooting Guide Files	Description
ACP_EAB_ts_guide	EAB system troubleshooting text file.
ACP_ILC_ts_guide	ILC system troubleshooting text file.
BRAKE_INTERFACE_ts_guide	Penalty & Cab Signal troubleshooting text file.
CAB_SIGNAL_ASPECTS_ts_guide	Cab Signal Monitor troubleshooting text file.
CDU_ts_guide	CDU troubleshooting text file.
CLASS_D_ts_guide	Communications Management Unit troubleshooting text file.
CPU_ts_guide	CPU troubleshooting text file.
DIO_ts_guide	DIO troubleshooting text file.
DISCRETE_INPUTS_ts_guide	Discrete Input troubleshooting text file.
EBI_ts_guide	EBI troubleshooting text file.
GPS_ts_guide	GPS troubleshooting text file.
HEAD_OF_TRAIN_ts_guide	Head-Of-Train troubleshooting text file.
IETMS_CHR_ts_guide	Crash Hardened recorder troubleshooting text file.
IOC_ts_guide	IOC troubleshooting text file.

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Troubleshooting Guide Files	Description
LIG_ts_guide	LIG troubleshooting text file.
LOCO_ID_ts_guide	Loco ID module troubleshooting text file.
LSI_DP_ts_guide	LSI DP troubleshooting text file.
LSI_EAB_ts_guide	LSI EAB troubleshooting text file.
LSI_ILC_ts_guide	LSI ILC troubleshooting text file.
RSM_ts_guide	RSM troubleshooting text file.

2.3 I-ETMS Railroads

Configuration File: `railroads.cfg`

The `railroads.cfg` file defines the participating interoperable railroads during system provisioning in order to *bootstrap* the system.

NOTE Any change to these parameters must be approved by all interoperating railroads.

The MDM server EMP address and default BOS EMP address are only used from this file until a BOSPolygon_RR.cfg file is downloaded for a railroad.

Table 2.3: railroad.cfg Parameters

RR	Parameter	Value
[ACEX]	<i>Possible Reporting Marks</i>	ACEX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	acex.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	acex.b:kes
	DEFAULT_BOS_EMP_ADDRESS	acex.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[AMTK]	<i>Possible Reporting Marks</i>	AMTK CDTX IDTX RNCX WDTX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	amtk.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:7002
	DEFAULT_KES_EMP_ADDRESS	amtk.b:kes
	DEFAULT_BOS_EMP_ADDRESS	amtk.b:cibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:7001
[ARR]	<i>Possible Reporting Marks</i>	ARR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	arr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	arr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	arr.b:1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[BNSF]	<i>Possible Reporting Marks</i>	BNSF BN ATSF BUGX EMD EMDX FURX HBT HLCX METX MNRX NREX RPRX SDRX

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RR	Parameter	Value
	DEFAULT_MDM_SERVER_EMP_ADDRESS	bnsf.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.254:11002
	DEFAULT_KES_EMP_ADDRESS	bnsf.b:kes
	DEFAULT_BOS_EMP_ADDRESS	bnsf.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.254:11000
[BRC]	<i>Possible Reporting Marks</i>	BRC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	brc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	brc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	brc.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[CFRC]	<i>Possible Reporting Marks</i>	CFRC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	cfrb.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	cfrb.b:kes
	DEFAULT_BOS_EMP_ADDRESS	cfrb.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[CLXX]	<i>Possible Reporting Marks</i>	CLXX AVR SWP OHIO DCR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	clxx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	clxx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	clxx.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[CN]	<i>Possible Reporting Marks</i>	CN BCOL BLE DMIR DWP EJE GTW IANR IC WC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	cn.b:mdm0
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8001
	DEFAULT_KES_EMP_ADDRESS	cn.b:kes
	DEFAULT_BOS_EMP_ADDRESS	cn.b:1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:8000
[CNR]	<i>Possible Reporting Marks</i>	CNR CNCA
	DEFAULT_MDM_SERVER_EMP_ADDRESS	cn.b:mdm0
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8001
	DEFAULT_KES_EMP_ADDRESS	cn.b:kes
	DEFAULT_BOS_EMP_ADDRESS	cn.b:1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:8000
[CPRS]	<i>Possible Reporting Marks</i>	CP CPRS CEFX SOO
	DEFAULT_MDM_SERVER_EMP_ADDRESS	cprs.b:mdm0
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8300
	DEFAULT_KES_EMP_ADDRESS	cprs.b:kes

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RR	Parameter	Value
	DEFAULT BOS EMP ADDRESS	cprs.b:ptcsrvr
	DEFAULT AG IP ADDRESS	10.255.255.20:8000
[CSAO]	<i>Possible Reporting Marks</i>	CSAO
	DEFAULT_MDM_SERVER_EMP_ADDRESS	csao.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8002
	DEFAULT_KES_EMP_ADDRESS	csao.b:kes
	DEFAULT BOS EMP ADDRESS	csao.b:gcibos
	DEFAULT AG IP ADDRESS	10.255.255.20:8001
[CSS]	<i>Possible Reporting Marks</i>	CSS
	DEFAULT_MDM_SERVER_EMP_ADDRESS	nicd.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	nicd.b:kes
	DEFAULT BOS EMP ADDRESS	nicd.b:ptcsrvr
	DEFAULT AG IP ADDRESS	10.255.255.20:11000
[CSXT]	<i>Possible Reporting Marks</i>	CSXT CON CEFX CLIN CREX FURX GCFX HATX HLCX INRD LLPX MPEX NHN PAL RLCX RPRX SCL
	DEFAULT_MDM_SERVER_EMP_ADDRESS	csxt.b:mb1
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8011
	DEFAULT_KES_EMP_ADDRESS	csxt.b:kes
	DEFAULT BOS EMP ADDRESS	csxt.b:cb1
	DEFAULT AG IP ADDRESS	10.255.255.20:8010
[FEC]	<i>Possible Reporting Marks</i>	FEC BLF
	DEFAULT_MDM_SERVER_EMP_ADDRESS	fec.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	fec.b:kes
	DEFAULT BOS EMP ADDRESS	fec.b:bos
	DEFAULT AG IP ADDRESS	10.255.255.20:11000
[FWWR]	<i>Possible Reporting Marks</i>	FWWR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	fwwr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	fwwr.b:kes
	DEFAULT BOS EMP ADDRESS	fwwr.b:bos
	DEFAULT AG IP ADDRESS	10.255.255.20:11000
[GNWR]	<i>Possible Reporting Marks</i>	GNWR PNWR LDRR UTAH AZER DGNO MNA

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RR	Parameter	Value
		KWT AGR AKDM LRWN CFNR IORY
	DEFAULT_MDM_SERVER_EMP_ADDRESS	gnwr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:2000
	DEFAULT_KES_EMP_ADDRESS	gnwr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	gnwr.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[IAIS]	<i>Possible Reporting Marks</i>	IAIS
	DEFAULT_MDM_SERVER_EMP_ADDRESS	iais.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	iais.b:kes
	DEFAULT_BOS_EMP_ADDRESS	iais.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[IANR]	<i>Possible Reporting Marks</i>	IANR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	ianr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	ianr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	ianr.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[IHB]	<i>Possible Reporting Marks</i>	IHB
	DEFAULT_MDM_SERVER_EMP_ADDRESS	ihb.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	ihb.b:kes
	DEFAULT_BOS_EMP_ADDRESS	ihb.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[JPBX]	<i>Possible Reporting Marks</i>	JPBX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	jpbx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	jpbx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	jpbx.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[KCS]	<i>Possible Reporting Marks</i>	KCS KCSM GMDX GWWR HLCX MRRC TFM
	DEFAULT_MDM_SERVER_EMP_ADDRESS	kcs.b:mbm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8206
	DEFAULT_KES_EMP_ADDRESS	kcs.b:kes
	DEFAULT_BOS_EMP_ADDRESS	kcs.b:1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:8205
[KCT]	<i>Possible Reporting Marks</i>	KCT
	DEFAULT_MDM_SERVER_EMP_ADDRESS	kct.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	kct.b:kes
	DEFAULT_BOS_EMP_ADDRESS	kct.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[LVRR]	<i>Possible Reporting Marks</i>	LVRR NSHR

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RR	Parameter	Value
	DEFAULT_MDM_SERVER_EMP_ADDRESS	lvrr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	lvrr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	lvrr.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[MARC]	<i>Possible Reporting Marks</i>	MARC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	marc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	marc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	marc.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[MBTA]	<i>Possible Reporting Marks</i>	MBTA
	DEFAULT_MDM_SERVER_EMP_ADDRESS	mbta.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	mbta.b:kes
	DEFAULT_BOS_EMP_ADDRESS	mbta.b:cibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[METX]	<i>Possible Reporting Marks</i>	METX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	metx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	metx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	mext.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[MNN]	<i>Possible Reporting Marks</i>	MNN
	DEFAULT_MDM_SERVER_EMP_ADDRESS	mnn.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	mnn.b:kes
	DEFAULT_BOS_EMP_ADDRESS	mnn.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[NCRC]	<i>Possible Reporting Marks</i>	NCRC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	ncrc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	ncrc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	ncrc.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[NCTC]	<i>Possible Reporting Marks</i>	NCTC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	nctc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:4322
	DEFAULT_KES_EMP_ADDRESS	nctc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	nctc.b:1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:4321
[NICD]	<i>Possible Reporting Marks</i>	NICD CSS
	DEFAULT_MDM_SERVER_EMP_ADDRESS	nicd.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	nicd.b:kes
	DEFAULT_BOS_EMP_ADDRESS	nicd.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[NJTR]	<i>Possible Reporting Marks</i>	NJTR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	njtr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	njtr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	njtr.b:gcibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000

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RR	Parameter	Value
[NMRX]	<i>Possible Reporting Marks</i>	NMRX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	nmx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	nmx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	nmx.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[NS]	<i>Possible Reporting Marks</i>	NS
	DEFAULT_MDM_SERVER_EMP_ADDRESS	ns.b:mdm1
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8002
	DEFAULT_KES_EMP_ADDRESS	ns.b:kes
	DEFAULT_BOS_EMP_ADDRESS	ns.b:b1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:8001
[NYSW]	<i>Possible Reporting Marks</i>	NYSW
	DEFAULT_MDM_SERVER_EMP_ADDRESS	nysw.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:7002
	DEFAULT_KES_EMP_ADDRESS	nysw.b:kes
	DEFAULT_BOS_EMP_ADDRESS	nysw.b:cibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:7001
[OMLX]	<i>Possible Reporting Marks</i>	OMLX NKCR
	DEFAULT_MDM_SERVER_EMP_ADDRESS	omlx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	omlx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	omlx.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[PJPB]	<i>Possible Reporting Marks</i>	PJPB JPBX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	pjpb.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	pjpb.b:kes
	DEFAULT_BOS_EMP_ADDRESS	pjpb.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[RJCC]	<i>Possible Reporting Marks</i>	RJCC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	rjcc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	rjcc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	rjcc.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[RTDC]	<i>Possible Reporting Marks</i>	RTDC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	rtdc.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:2000
	DEFAULT_KES_EMP_ADDRESS	rtdc.b:kes
	DEFAULT_BOS_EMP_ADDRESS	rtdc.b:b1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:57500
[RTDZ]	<i>Possible Reporting Marks</i>	RTDZ RTDC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	rtdz.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:2000
	DEFAULT_KES_EMP_ADDRESS	rtdz.b:kes
	DEFAULT_BOS_EMP_ADDRESS	rtdz.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:57500
[SCAX]	<i>Possible Reporting Marks</i>	SCAX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	scax.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	scax.b:kes

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RR	Parameter	Value
	DEFAULT_BOS_EMP_ADDRESS	scax.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[SCIH]	<i>Possible Reporting Marks</i>	SCIH
	DEFAULT_MDM_SERVER_EMP_ADDRESS	scih.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	scih.b:kes
	DEFAULT_BOS_EMP_ADDRESS	scih.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[SDRX]	<i>Possible Reporting Marks</i>	SDRX AMTK
	DEFAULT_MDM_SERVER_EMP_ADDRESS	bnsf.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.254:11002
	DEFAULT_KES_EMP_ADDRESS	bnsf.b:kes
	DEFAULT_BOS_EMP_ADDRESS	bnsf.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.254:11000
[SEPA]	<i>Possible Reporting Marks</i>	SEPA
	DEFAULT_MDM_SERVER_EMP_ADDRESS	sepa.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:8002
	DEFAULT_KES_EMP_ADDRESS	sepa.b:kes
	DEFAULT_BOS_EMP_ADDRESS	sepa.b:gcibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:8001
[SFRV]	<i>Possible Reporting Marks</i>	SFRV TRCX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	sfrv.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	sfrv.b:kes
	DEFAULT_BOS_EMP_ADDRESS	sfrv.b:ptcsrvr
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[ST]	<i>Possible Reporting Marks</i>	ST BM MEC PTM
	DEFAULT_MDM_SERVER_EMP_ADDRESS	st.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	st.b:kes
	DEFAULT_BOS_EMP_ADDRESS	st.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[TCWR]	<i>Possible Reporting Marks</i>	TCWR RRVW
	DEFAULT_MDM_SERVER_EMP_ADDRESS	tcwr.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	tcwr.b:kes
	DEFAULT_BOS_EMP_ADDRESS	tcwr.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[TMBL]	<i>Possible Reporting Marks</i>	TMBL
	DEFAULT_MDM_SERVER_EMP_ADDRESS	tmb1.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	tmb1.b:kes
	DEFAULT_BOS_EMP_ADDRESS	tmb1.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[TRE]	<i>Possible Reporting Marks</i>	TRE TEXR GVRX SLRC
	DEFAULT_MDM_SERVER_EMP_ADDRESS	tre.b:mdm

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RR	Parameter	Value
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	tre.b:kes
	DEFAULT_BOS_EMP_ADDRESS	tre.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[TRRA]	Possible Reporting Marks	TRRA
	DEFAULT_MDM_SERVER_EMP_ADDRESS	up.b:trramdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11003
	DEFAULT_KES_EMP_ADDRESS	up.b:kes
	DEFAULT_BOS_EMP_ADDRESS	up.b:trra
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11004
[TXPF]	Possible Reporting Marks	TXPF FXE
	DEFAULT_MDM_SERVER_EMP_ADDRESS	txpf.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	txpf.b:kes
	DEFAULT_BOS_EMP_ADDRESS	txpf.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[UP]	Possible Reporting Marks	UP ALS CEFX CMC CNW DRGW FURX GECX GMTX HLCX LLPX METX PTRA SP SSW UPB UPS UPT UPY
	DEFAULT_MDM_SERVER_EMP_ADDRESS	up.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11003
	DEFAULT_KES_EMP_ADDRESS	up.b:kes
	DEFAULT_BOS_EMP_ADDRESS	up.b:p1
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11004
[VREX]	Possible Reporting Marks	VREX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	vrex.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:7002
	DEFAULT_KES_EMP_ADDRESS	vrex.b:kes
	DEFAULT_BOS_EMP_ADDRESS	vrex.b:cibos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:7001
[WAMX]	Possible Reporting Marks	WAMX
	DEFAULT_MDM_SERVER_EMP_ADDRESS	wamx.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002
	DEFAULT_KES_EMP_ADDRESS	wamx.b:kes
	DEFAULT_BOS_EMP_ADDRESS	wamx.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000
[WE]	Possible Reporting Marks	WE
	DEFAULT_MDM_SERVER_EMP_ADDRESS	we.b:mdm
	DEFAULT_MDM_IP_ADDRESS	10.255.255.20:11002

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RR	Parameter	Value
	DEFAULT_KES_EMP_ADDRESS	we.b:kes
	DEFAULT_BOS_EMP_ADDRESS	we.b:bos
	DEFAULT_AG_IP_ADDRESS	10.255.255.20:11000

2.4 I-ETMS QoS Levels

Configuration File: `ietms_qos.cfg`

The `ietms_qos.cfg` file contains the Quality of Service (QoS) and Time to Live (TTL) settings for each Office-Locomotive, WSRS-Locomotive, WIU-Locomotive, and IVOC-Locomotive message. Every message has three priority settings in the configuration file. When the message has only one defined priority value, all three settings are set to that value. The TTL setting for each message associated with TBC101 must be less than TBC101 in section 2.1 TBC Common Parameters.

The QoS values defined for the Office-Locomotive and WSRS-Locomotive messages in the QoS parameters table are captured in the I-ETMS Protect PTC Office – Locomotive Segment ICD.

The QoS values defined for the WIU-Locomotive messages in the QoS parameters table are captured in the ITC Wayside-Locomotive ICD.

The QoS values defined in the IVOC-Locomotive messages in the QoS parameters table are captured in the IVOC-TMC ICD.

NOTE Any change to these parameters must be approved by all interoperating railroads.

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Table 2.4: I-ETMS QoS Parameters

Msg#	Msg_name	priority_at _fast_poll Range	priority_at _medium_ poll Range	priority_at _slow_poll Range	class Range	network_preference Range	special_ handling Range	service_rqst Range	time_to_live (seconds) Range
[02000]	Verify Employee Info Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02001]	Request Train ID List	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02003]	Selected Train ID	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02005]	Configuration Version List Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02007]	Request Train Subdivision/District List	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02008]	Confirmation of Train Subdivision/District List	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02010]	Locomotive System State	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02011]	Departure Test Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02020]	Poll Registration	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02022]	Request Current Dataset List	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02030]	Request Train Consist	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02031]	Confirmation of Train Consist	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02032]	Onboard Train Consist	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02040]	Confirmation of Crew Acknowledgement of Bulletin	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02041]	Request Bulletin Dataset	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02042]	Confirmation of Bulletin Dataset	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800

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Msg#	Msg_name	priority_at _fast_poll Range	priority_at _medium_ poll Range	priority_at _slow_poll Range	class Range	network_preference Range	special handling Range	service_rqst Range	time_to_live (seconds) Range
[02043]	Confirmation of Bulletin Cancellation	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02050]	Crew Authority Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02051]	Request Movement Authority Dataset	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02052]	Confirmation of Movement Authority	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02053]	Confirmation of Movement Authority Void	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02054]	EMT/PSS Authority Used Notification	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02060]	Confirmation of Crew Message	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02061]	Crew Acknowledgement of Mandatory Directive Status	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02062]	Confirmation of Bulletin Sequence	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02070]	Onboard Violation Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02071]	Confirmation of Violation Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02072]	Onboard Violation Cleared	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02073]	Confirmation of Violation Cleared	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02080]	Locomotive Position Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02081]	Locomotive System Fault Summary Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02082]	PTC Interaction	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02083]	Enforcement	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800

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Msg#	Msg_name	priority_at _fast_poll Range	priority_at _medium_ poll Range	priority_at _slow_poll Range	class Range	network_preference Range	special_ handling Range	service_rqst Range	time_to_live (seconds) Range
	Warning/Braking Notification								
[02084]	Emergency Brake Application Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02085]	Train Handling Exception Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02086]	Acknowledge Conditional Authority Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02087]	Locomotive Fault Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02088]	Onboard Component Configuration Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02100]	Client File Set List	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02101]	File Info Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02102]	File Part Checksum Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02103]	File Data Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02104]	File Complete	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02105]	File Download Failed	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02120]	File Info	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02121]	No File	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02122]	File Data	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02500]	WSRS Subscribe	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02501]	WSRS Unsubscribe	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[5200]	Beacon Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[5201]	Get WIU Status	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02820]	IVOC Poll Registration	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02851]	IVOC Request Track Block Authority	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02852]	IVOC Confirmation	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800

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Msg#	Msg_name	priority_at _fast_poll Range	priority_at _medium_ poll Range	priority_at _slow_poll Range	class Range	network_preference Range	special_ handling Range	service_rqst Range	time_to_live (seconds) Range
	of Track Block Authority								
[02853]	IVOC Confirmation of Void Track Block Authority	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02859]	IVOC Confirmation of Wait Location Update	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02861]	Crew Acknowledgement of Mandatory Directive	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02871]	IVOC Confirmation of Authority Violation Report	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[02880]	IVOC On-Board Status Message	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[06025]	Text Message	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[06026]	Recipient List Request	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800
[06027]	Confirmation of Text Message	0-7	0-7	0-7	0-7	0-7	0-15	0-7	0-64800

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2.5 Energy Management Safety Messages

Configuration File: em_safety_msgs.cfg

The TBC common messages contained in the `em_safety_msgs.cfg` file are configured for interoperability and used by all owning, employing, and operating railroads.

NOTE Any change to these parameters must be approved by all interoperating railroads.

All EM Safety Messages will be referenced by number for display.

All EM Safety Messages shall include a linefeed character between lines of text.

Insertion Characters provided by the requesting message shall be inserted as follows:

- %d = Countdown timer, displayed as a 2 digit decimal number (counts as 2 characters) (A countdown timer can be used in Non-Enumerated messages as well)
- %t = Throttle Notch, displayed as 'Nx', where x is the notch value supplied (Max value 8) (counts as 2 characters)
- %b = Dynamic Braking Notch, displayed as 'Dx', where x is the braking notch supplied (Max value 8) (counts as 2 characters)
- %p = Braking Pressure, displayed as a 2 character decimal number (Max value 99) (counts as 2 characters)
- %i = Integer Insertion, displayed as a single digit decimal number (counts as 1 character)

Table 2.5: Energy Management Safety Messages

Message Number	Safety Message
0	MANUAL CONTROL NEEDED NOW! %d
1	WAITING FOR MATCH... %d
2	TIMED OUT! GOING TO IDLE
3	WAITING FOR N8 %d
4	WAITING FOR CONFIRMATION %d
5	PLACE THROTTLE IN %t NOW
6	PLACE THROTTLE IN %b NOW
7	REDUCE BPP TO %p IN %d
8	CREW NOW HAS MANUAL CONTROL
9	WAITING FOR MATCH...
128	SYSTEM FAULT AUTO DISENGAGING MANUAL CONTROL IN %d SECONDS
129	AUTO DISENGAGING MANUAL CONTROL IN %d SECONDS
130	TH/DB MATCH FAIL AUTO DISENGAGING

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Message Number	Safety Message
	MANUAL CONTROL IN %d SECONDS
131	TH/DB MOVED FROM STARTING POINT AUTO DISENGAGED
132	AUTO DISENGAGE NOT VERIFIED AUTO DISENGAGING MANUAL CONTROL IN %d SECONDS

Note: 2 line display messages shall use numbers 0 through 127 while 5 line display messages shall use numbers 128 through 255.

2.6 Train Restriction Types

Configuration File: train_restriction_types.cfg

The train_restriction_types.cfg file contains textual descriptions for the train restriction types that are common for all interoperable railroads. The numeric value assigned to each train restriction type corresponds to the value assigned in the summary consist messages exchanged with the back office and to the value assigned in the track data for the train restriction type.

NOTE Any change to these parameters must be approved by all interoperating railroads.

The format of the train_restriction_types.cfg file is shown below:

```
[TRAIN_RESTRICTION_TYPES]
```

```
#for each train restriction type
```

```
<train restriction type number> <train restriction type description>
```

Any train_restriction_types.cfg file that defines more than one description for a train restriction type number is considered invalid.

Any train_restriction_types.cfg file that contains a train restriction type description longer than 30 characters is considered invalid.

Any train_restriction_types.cfg file that defines a train restriction type description for a type number outside the range of 1 – 16 is considered invalid.

Any train_restriction_types.cfg file that defines a train restriction type without a corresponding description is considered invalid.

Table 2.6: Common Train Restriction Types

Number	Description
1	KEY TRAIN
2	HIGH HAZARD FLAMMABLE TRAIN

2.7 Navigation Module

Configuration File: nav_module.cfg

The nav_module.cfg file contains on-board segment configuration information related to the interface with the navigation modules.

The public keys used as part of the navigation module authentication process are assigned to each navigation module type within this file by referencing a .cfg file that contains the formatted public key.

The format of the nav_module.cfg file is shown below:

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[public key table]

for each public key index

<public key index> <public key filename>.cfg

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/// 3 Railroad Configuration Fileset

The railroad configurable filesets include:

- params_RR.cfg
- params_THE_RR.cfg
- BOSPolygon_RR.cfg
- params_comm_RR.cfg
- params_vlan_RR.cfg
- license_RR.cfg
- train_restriction_types_RR.cfg

All railroad configurable files must be included in the fileset, with the exception of the params_vlan_RR.cfg fileset, are required in order to operate the On-board segment. Should the on-board be unable to locate or verify a required .cfg file, the System Configuration check during initialization will not pass.

3.1 Railroad TBC Values and Functional Parameters

Configuration File: **params_RR.cfg**

NOTE The **RR** characters in the actual configuration files are replaced with the individual railroad SCAC code.

Each version of the params_RR.cfg file contains railroad specific parameters that are configured by the individual ITC member railroads. These parameters include TBC values and functional parameters that are used to enable or disable of on-board functions. Changes to these parameters are made at the discretion of the railroad.

Note that for railroads operating in metric, all parameters representing speeds must be entered as kilometers per hour within the ranges defined in the table below for applicable parameters.

Table 3.1: Railroad TBC Values

TBC#	Description	Units	Valid Range (min-max)	Railroad
TBC19	Clearance Number Digits <i>The number of digits to be displayed on the clearance number entry screen.</i>	n/a	4-8	Operating
TBC46	Employee ID Digits <i>The number of digits to be displayed on the employee ID entry screen.</i>	digit	4-8	Employing
TBC47	Employee PIN Digits <i>The number of digits to be displayed on the employee PIN entry screen.</i>	digit	4-8	Employing
TBC66	Horn Protection Activation Time Threshold <i>The minimum time away from a crossing that the on-board segment will activate horn protection when the train speed is at or below Maximum Speed for Time Based Horn Protection.</i>	seconds	0-180	Operating railroad of the grade crossing
TBC67	Horn Protection Minimum Speed <i>When train speed is less than Horn Protection Minimum Speed, the on-board</i>	mph	0-125	Operating railroad of the grade
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>system activation of the horn will cease.</i>			crossing
TBC137	Restricted Speed <i>Maximum speed allowed under Restricted Speed rules, used on track where the country of operation is not Canada</i>	mph	1-20	Operating
		km/h	1-32	
TBC140	SG-3 Freight Target Speed <i>Maximum allowed freight speed for operation under signal group 3</i>	mph	5-125	Operating railroad of the signal
		km/h	8-201	
TBC141	SG-3 Passenger Target Speed <i>Maximum allowed passenger speed for operation under signal group 3</i>	mph	5-125	Operating railroad of the signal
		km/h	8-201	
TBC158	Switch Prompting Distance	feet	1-10000	Operating railroad of the switch
TBC222	Departure Test Full Service Threshold <i>The amount by which the front brake pipe pressure must be reduced in order to pass the brake test portion of the departure test.</i>	psi	0-40	Owning
TBC265	WSRS Backup Enabled subscription distance <i>The distance between the train's leading edge and a wayside device at which the on-board segment will subscribe for status via WSRS if the WIU monitoring that device is designated as WSRS Backup in the track data and the on-board segment is not receiving status for that device.</i>	feet	0-42,240	Operating railroad of the wayside device
TBC267	ITC EMP Time Message Broadcast Period <i>Configures the rate at which the ITC EMP Time Message is broadcast as described in the ITC Time and Location ICD</i>	seconds	0-4095	Owning
TBC268	ITC EMP Location Message Broadcast Period <i>Configures the rate at which the ITC EMP Location Message is broadcast as described in the ITC Time and Location ICD</i>	seconds	0-4095	Owning
TBC281	Train Speed Margin Prior To Delay In Block <i>On track that is signaled in the train's direction of movement and either requires form based authority in at least one direction or is designated Yard Limits the train speed must be at least TBC281 + TBC282 mph since the train entered the signal block in order to invoke the on-board segment's delayed in block functionality</i>	mph	0-125	Operating railroad of the signal controlling movement in the block
		km/h	0-201	
TBC282	Train Speed To Invoke Delay In Block <i>On track that is signaled in the train's direction of movement and either requires form based authority in at least one direction or is designated Yard Limits the on-board segment's delayed in block functionality is invoked if the train speed drops to TBC282 mph or below</i>	mph	0-125	Operating railroad of the signal controlling movement in the block
		km/h	0-201	
TBC283	Delayed In Block Target Speed <i>When the on-board segment's delayed in</i>	mph	1-125	Operating railroad of the
		km/h	1-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>block functionality is invoked on track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the speed assigned to the delayed in block target</i> <i>See also CFG16 – If CFG16 is enabled and the country of operation is not Canada, the target speed is set to the lower of TBC137 mph or TBC283 mph. If CFG16 is enabled and the country of operation is Canada, the target speed is set to the lower of TBC479 or TBC283 mph.</i>			signal controlling movement in the block
TBC284	Pass Signal At Stop In ABS Target Distance <i>On track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the distance in advance of the signal at stop that the on-board segment generates a target after the crew indicates that they are unable to contact the dispatcher and can see the signal at stop.</i> <i>See also CFG21</i>	feet	0-10000	Operating railroad of the signal
TBC285	Pass Signal At Stop In ABS Prompt Distance <i>On track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the distance from the target TBC284 feet in advance of the signal at stop at which the on-board segment prompts for authority to proceed to the next signal. This distance should be less than TBC284 to ensure that that the leading wheels of the locomotive have passed the signal at stop.</i> <i>See also CFG21</i>	feet	0-10000	Operating railroad of the signal
TBC286	Pass Signal At Stop In ABS Prompt Timeout <i>On track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the period after which the on-board segment prompts for authority to proceed to the next signal if the leading edge of the train is within TBC285 feet of the target TBC284 feet in advance of the signal at stop</i> <i>See also CFG21</i>	seconds	0-300	Operating railroad of the signal
TBC290	Work Zone Display Time <i>Defines how long prior to the effective time of a work zone the on-board segment will generate targets for that work zone</i> <i>See also CFG23</i>	seconds	0-3600	Operating railroad issuing the work zone bulletin
TBC308	Freight Compliant Speed for Unintended State Change <i>The speed enforced by the on-board segment upon an "unintended" transition from the ACTIVE state to the DISENGAGED, CUT OUT or FAILED state for a freight or</i>	mph	1-125	Operating
		km/h	1-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>intermodal train.</i>			
TBC315	Transition to Active Delay <i>The period that the on-board segment will wait between establishing a locomotive track location and transitioning from the DISENGAGED state to the ACTIVE state in order to allow time to obtain status from wayside devices in advance of the leading edge of the train on the train's calculated route.</i>	seconds	4-18	Operating
TBC316	Subdivision Speed for Train Restriction Type 1 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 1 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC316.</i>	mph km/h	0-125 0-201	Operating
TBC317	Subdivision Speed for Train Restriction Type 2 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 2 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC317.</i>	mph km/h	0-125 0-201	Operating
TBC318	Subdivision Speed for Train Restriction Type 3 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 3 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC318.</i>	mph km/h	0-125 0-201	Operating
TBC319	Subdivision Speed for Train Restriction Type 4 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 4 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC319.</i>	mph km/h	0-125 0-201	Operating
TBC320	Subdivision Speed for Train Restriction Type 5 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 5 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC320.</i>	mph km/h	0-125 0-201	Operating
TBC321	Subdivision Speed for Train Restriction Type 6 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 6 and the speed</i>	mph km/h	0-125 0-201	Operating

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC321.</i>			
TBC322	Subdivision Speed for Train Restriction Type 7 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 7 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC322.</i>	mph	0-125	Operating
		km/h	0-201	
TBC323	Subdivision Speed for Train Restriction Type 8 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 8 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC323.</i>	mph	0-125	Operating
		km/h	0-201	
TBC324	Subdivision Speed for Train Restriction Type 9 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 9 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC324.</i>	mph	0-125	Operating
		km/h	0-201	
TBC325	Subdivision Speed for Train Restriction Type 10 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 10 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC325.</i>	mph	0-125	Operating
		km/h	0-201	
TBC326	Subdivision Speed for Train Restriction Type 11 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 11 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC326.</i>	mph	0-125	Operating
		km/h	0-201	
TBC327	Subdivision Speed for Train Restriction Type 12 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 12 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC327.</i>	mph	0-125	Operating
		km/h	0-201	
TBC328	Subdivision Speed for Train Restriction Type 13 <i>When the on-board segment generates a speed restriction target for a subdivision-wide</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>train restriction of type 13 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC328.</i>			
TBC329	Subdivision Speed for Train Restriction Type 14 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 14 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC329.</i>	mph	0-125	Operating
		km/h	0-201	
TBC330	Subdivision Speed for Train Restriction Type 15 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 15 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC330.</i>	mph	0-125	Operating
		km/h	0-201	
TBC331	Subdivision Speed for Train Restriction Type 16 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 16 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC331.</i>	mph	0-125	Operating
		km/h	0-201	
TBC332	Subdivision Speed for Train Restriction Type 17 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 17 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC332.</i>	mph	0-125	Operating
		km/h	0-201	
TBC333	Subdivision Speed for Train Restriction Type 18 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 18 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC333.</i>	mph	0-125	Operating
		km/h	0-201	
TBC334	Subdivision Speed for Train Restriction Type 19 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 19 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC334.</i>	mph	0-125	Operating
		km/h	0-201	
TBC335	Subdivision Speed for Train Restriction Type 20 <i>When the on-board segment generates a</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>speed restriction target for a subdivision-wide train restriction of type 20 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC335.</i>			
TBC336	Subdivision Speed for Train Restriction Type 21 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 21 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC336.</i>	mph km/h	0-125 0-201	Operating
TBC337	Subdivision Speed for Train Restriction Type 22 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 22 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC337.</i>	mph km/h	0-125 0-201	Operating
TBC338	Subdivision Speed for Train Restriction Type 23 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 23 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC338.</i>	mph km/h	0-125 0-201	Operating
TBC339	Subdivision Speed for Train Restriction Type 24 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 24 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC339.</i>	mph km/h	0-125 0-201	Operating
TBC340	Subdivision Speed for Train Restriction Type 25 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 25 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC340.</i>	mph km/h	0-125 0-201	Operating
TBC341	Subdivision Speed for Train Restriction Type 26 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 26 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC341.</i>	mph km/h	0-125 0-201	Operating
TBC342	Subdivision Speed for Train Restriction Type 27	mph km/h	0-125 0-201	Operating

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 27 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC342.</i>			
TBC343	Subdivision Speed for Train Restriction Type 28 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 28 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC343.</i>	mph	0-125	Operating
		km/h	0-201	
TBC344	Subdivision Speed for Train Restriction Type 29 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 29 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC344.</i>	mph	0-125	Operating
		km/h	0-201	
TBC345	Subdivision Speed for Train Restriction Type 30 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 30 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC345.</i>	mph	0-125	Operating
		km/h	0-201	
TBC346	Subdivision Speed for Train Restriction Type 31 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 31 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC346.</i>	mph	0-125	Operating
		km/h	0-201	
TBC347	Subdivision Speed for Train Restriction Type 32 <i>When the on-board segment generates a speed restriction target for a subdivision-wide train restriction of type 32 and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC347.</i>	mph	0-125	Operating
		km/h	0-201	
TBC348	Area-Specific Speed for Train Restriction Type 1 <i>When the on-board segment generates a speed restriction target for a train restriction of type 1 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC348.</i>	mph	0-125	Operating
		km/h	0-201	
TBC349	Area-Specific Speed for Train Restriction	mph	0-125	Operating

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	Type 2 <i>When the on-board segment generates a speed restriction target for a train restriction of type 2 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC349.</i>	km/h	0-201	
TBC350	Area-Specific Speed for Train Restriction Type 3 <i>When the on-board segment generates a speed restriction target for a train restriction of type 3 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC350.</i>	mph	0-125	Operating
		km/h	0-201	
TBC351	Area-Specific Speed for Train Restriction Type 4 <i>When the on-board segment generates a speed restriction target for a train restriction of type 4 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC351.</i>	mph	0-125	Operating
		km/h	0-201	
TBC352	Area-Specific Speed for Train Restriction Type 5 <i>When the on-board segment generates a speed restriction target for a train restriction of type 5 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC352.</i>	mph	0-125	Operating
		km/h	0-201	
TBC353	Area-Specific Speed for Train Restriction Type 6 <i>When the on-board segment generates a speed restriction target for a train restriction of type 6 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC353.</i>	mph	0-125	Operating
		km/h	0-201	
TBC354	Area-Specific Speed for Train Restriction Type 7 <i>When the on-board segment generates a speed restriction target for a train restriction of type 7 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC354.</i>	mph	0-125	Operating
		km/h	0-201	
TBC355	Area-Specific Speed for Train Restriction Type 8 <i>When the on-board segment generates a speed restriction target for a train restriction of type 8 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC355.</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
TBC356	Area-Specific Speed for Train Restriction Type 9 <i>When the on-board segment generates a speed restriction target for a train restriction of type 9 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC356.</i>	mph	0-125	Operating
		km/h	0-201	
TBC357	Area-Specific Speed for Train Restriction Type 10 <i>When the on-board segment generates a speed restriction target for a train restriction of type 10 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC357.</i>	mph	0-125	Operating
		km/h	0-201	
TBC358	Area-Specific Speed for Train Restriction Type 11 <i>When the on-board segment generates a speed restriction target for a train restriction of type 11 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC358.</i>	mph	0-125	Operating
		km/h	0-201	
TBC359	Area-Specific Speed for Train Restriction Type 12 <i>When the on-board segment generates a speed restriction target for a train restriction of type 12 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC359.</i>	mph	0-125	Operating
		km/h	0-201	
TBC360	Area-Specific Speed for Train Restriction Type 13 <i>When the on-board segment generates a speed restriction target for a train restriction of type 13 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC360.</i>	mph	0-125	Operating
		km/h	0-201	
TBC361	Area-Specific Speed for Train Restriction Type 14 <i>When the on-board segment generates a speed restriction target for a train restriction of type 14 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC361.</i>	mph	0-125	Operating
		km/h	0-201	
TBC362	Area-Specific Speed for Train Restriction Type 15 <i>When the on-board segment generates a speed restriction target for a train restriction of type 15 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>segment sets the target speed to TBC362.</i>			
TBC363	Area-Specific Speed for Train Restriction Type 16 <i>When the on-board segment generates a speed restriction target for a train restriction of type 16 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC363.</i>	mph	0-125	Operating
		km/h	0-201	
TBC364	Area-Specific Speed for Train Restriction Type 17 <i>When the on-board segment generates a speed restriction target for a train restriction of type 17 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC364.</i>	mph	0-125	Operating
		km/h	0-201	
TBC365	Area-Specific Speed for Train Restriction Type 18 <i>When the on-board segment generates a speed restriction target for a train restriction of type 18 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC365.</i>	mph	0-125	Operating
		km/h	0-201	
TBC366	Area-Specific Speed for Train Restriction Type 19 <i>When the on-board segment generates a speed restriction target for a train restriction of type 19 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC366.</i>	mph	0-125	Operating
		km/h	0-201	
TBC367	Area-Specific Speed for Train Restriction Type 20 <i>When the on-board segment generates a speed restriction target for a train restriction of type 20 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC367.</i>	mph	0-125	Operating
		km/h	0-201	
TBC368	Area-Specific Speed for Train Restriction Type 21 <i>When the on-board segment generates a speed restriction target for a train restriction of type 21 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC368.</i>	mph	0-125	Operating
		km/h	0-201	
TBC369	Area-Specific Speed for Train Restriction Type 22 <i>When the on-board segment generates a speed restriction target for a train restriction of type 22 with limits defined in the track data and the speed designated in the track data</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>for that restriction is 0 mph, the on-board segment sets the target speed to TBC369.</i>			
TBC370	Area-Specific Speed for Train Restriction Type 23 <i>When the on-board segment generates a speed restriction target for a train restriction of type 23 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC370.</i>	mph	0-125	Operating
		km/h	0-201	
TBC371	Area-Specific Speed for Train Restriction Type 24 <i>When the on-board segment generates a speed restriction target for a train restriction of type 24 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC371.</i>	mph	0-125	Operating
		km/h	0-201	
TBC372	Area-Specific Speed for Train Restriction Type 25 <i>When the on-board segment generates a speed restriction target for a train restriction of type 25 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC372.</i>	mph	0-125	Operating
		km/h	0-201	
TBC373	Area-Specific Speed for Train Restriction Type 26 <i>When the on-board segment generates a speed restriction target for a train restriction of type 26 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC373.</i>	mph	0-125	Operating
		km/h	0-201	
TBC374	Area-Specific Speed for Train Restriction Type 27 <i>When the on-board segment generates a speed restriction target for a train restriction of type 27 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC374.</i>	mph	0-125	Operating
		km/h	0-201	
TBC375	Area-Specific Speed for Train Restriction Type 28 <i>When the on-board segment generates a speed restriction target for a train restriction of type 28 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC375.</i>	mph	0-125	Operating
		km/h	0-201	
TBC376	Area-Specific Speed for Train Restriction Type 29 <i>When the on-board segment generates a speed restriction target for a train restriction of type 29 with limits defined in the track data</i>	mph	0-125	Operating
		km/h	0-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC376.</i>			
TBC377	Area-Specific Speed for Train Restriction Type 30 <i>When the on-board segment generates a speed restriction target for a train restriction of type 30 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC377.</i>	mph	0-125	Operating
		km/h	0-201	
TBC378	Area-Specific Speed for Train Restriction Type 31 <i>When the on-board segment generates a speed restriction target for a train restriction of type 31 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC378.</i>	mph	0-125	Operating
		km/h	0-201	
TBC379	Area-Specific Speed for Train Restriction Type 32 <i>When the on-board segment generates a speed restriction target for a train restriction of type 32 with limits defined in the track data and the speed designated in the track data for that restriction is 0 mph, the on-board segment sets the target speed to TBC379.</i>	mph	0-125	Operating
		km/h	0-201	
TBC402	Passenger Compliant Speed for Unintended State Change <i>The speed enforced by the on-board segment upon an “unintended” transition from the ACTIVE state to the DISENGAGED, CUT OUT or FAILED state for a passenger, commuter, high speed passenger, or tilt train.</i>	mph	1-125	Operating
		km/h	1-201	
TBC415	MTEA Speed <i>The speed enforced for Main Line Track Exception Addendum when the on-board segment is Disengaged and not navigating.</i>	mph	1-125	Operating
		km/h	1-201	
TBC416	Horn Test Long Duration <i>The number of seconds the horn will be commanded to blow during the long duration horn test. If TBC416 is set to 1, the Long Duration Horn Test button will not be displayed.</i>	seconds	1-20	Owning
TBC417	Freight Shoving Move Speed Restriction <i>Maximum speed allowed for a train of type freight or intermodal when shoving cars. See also CFG36.</i>	mph	1-125	Operating
		km/h	1-201	
TBC418	Passenger Shoving Move Speed Restriction <i>Maximum speed allowed for a train of type passenger, commuter, high speed passenger, or tilt train when shoving cars. See also CFG36.</i>	mph	1-125	Operating
		km/h	1-201	
TBC421	Timeout to Remove Last Known Signal Target from Memory	minutes	1-60	Operating railroad where

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	When TBC421 minutes have elapsed since the on-board segment stored a signal target at a transition to Initializing, Cut Out or Restricted state and the locomotive track location and direction of movement are still unknown, the on-board segment will remove the retained signal target from memory. This prevents the on-board segment from retaining signal target status beyond, for example, the expected length of a crew change. Note that TBC421 is only in effect when CFG29 is enabled for the operating railroad where the leading edge of the train was located at the time of the state transition to Initializing, Cut Out or Restricted.			signal is located
TBC424	Obtain Switch Status Enforcement Delay The enforcement delay set by the on-board segment in the ACTIVE state when generating a target for a monitored switch with unknown status that is added to the train's calculated route in order to allow time to obtain status from that device prior to braking for that target.	seconds	0-5	Operating
TBC425	Obtain Signal Status Enforcement Delay The enforcement delay set by the on-board segment in the ACTIVE state when generating a target for a monitored signal with unknown status that is added to the train's calculated route in order to allow time to obtain status from that device prior to braking for that target.	seconds	0-5	Operating
TBC426	Zero Speed Signal Target Enforcement Delay The enforcement delay set by the on-board segment on any existing zero (0) mph unknown signal target when the on-board segment state transitions from DISENGAGED to ACTIVE.	seconds	0-180	Operating
TBC427	Maximum Antenna Offset Error The maximum amount of error allowed in the values entered for the antenna offset from centerline and antenna offset from front knuckle in the Loco ID	inches	1-8	Owning
TBC430	Serial Cab Signal System Degraded State Change Enforcement Delay The period for which the on-board segment will warn before initiating a full-service brake application if the train speed is above TBC435/TBC436 mph following a Cab Signal System State Change from the "Higher Speed Operation Enabled" Cab Signal Status when the locomotive is located within serial interface cab signal track.	seconds	0-180	Operating
TBC431	Freight Compliant Speed for Unequipped for Serial Cab Signal System	mph	1-125	Operating
		km/h	1-201	

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TBC#	Description	Units	Valid Range (min-max)	Railroad
	<i>The speed enforced for a freight or intermodal train when the locomotive is located within serial interface cab signal track and does NOT have a Serial Cab Signal System Interface.</i>			
TBC432	Passenger Compliant Speed for Unequipped for Serial Cab Signal System <i>The speed enforced for a passenger, commuter, high speed passenger, or tilt train when the locomotive is located within serial interface cab signal track and does NOT have a Serial Cab Signal System Interface.</i>	mph	1-125	Operating
		km/h	1-201	
TBC433	Excessive Serial Cab Signal System Status Transitions <i>The number of times the Cab Signal Status can transition between "Higher-Speed Operation Enabled" to "Higher-Speed Operation Disabled" before being considered excessive.</i>	n/a	2-10	Operating
TBC434	Excessive Serial Cab Signal System Status Transitions Duration <i>The number of seconds for the monitoring period of the Cab Signal Status for excessive transitions from "Higher-Speed Operation Enabled" to "Higher-Speed Operation Disabled".</i>	seconds	30-600	Operating
TBC435	Freight Compliant Speed for Degraded Serial Cab Signal System <i>The speed enforced by the on-board segment upon a transition from the "Higher Speed Operation Enabled" Cab Signal Status for a freight or intermodal train when the locomotive is located within serial interface cab signal track and does have a Serial Cab Signal System Interface.</i>	mph	1-125	Operating
		km/h	1-201	
TBC436	Passenger Compliant Speed for Degraded Serial Cab Signal System <i>The speed enforced by the on-board segment upon a transition from the "Higher Speed Operation Enabled" Cab Signal Status for a passenger, commuter, high speed passenger, or tilt train when the locomotive is located within serial interface cab signal track and does have a Serial Cab Signal System Interface.</i>	mph	1-125	Operating
		km/h	1-201	
TBC440	EM Recovery Message Display Time <i>Time to display the EM Recovery message after EM Communication Loss is declared.</i>	seconds	10-300	Owning
TBC441	Stale EM Display Timeout <i>Time to clear all EM Advisory displays after EM Communication Loss is declared and is still uncleared.</i>	seconds	15 - 30	Owning
TBC444	Data Non-Comm Area Prompt Display Time <i>Maximum time a prompt related to non-comm area of type Data entry/exit will be displayed.</i>	seconds	0-600	Operating

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TBC#	Description	Units	Valid Range (min-max)	Railroad
TBC445	WIUStatus Message Age Limit in Data Non-Comm Area <i>When the locomotive is located within a Data Non-Comm Area, the onboard segment will maintain the last valid WIU message received for up to TBC445 seconds before setting any associated wayside status to unknown.</i>	seconds	1-60	Operating
TBC446	Max Odometer	miles	0-620	Owning
TBC447	Automatic Parking Set Time <i>The number of seconds the train has been stopped and the reverser has been centered before the Park Function will be enabled.</i>	seconds	0-120	Owning
TBC448	Consist Length Tolerance for Train Dynamics <i>The percentage of the consist Total Length that the train is expected to change due to train dynamics.</i>	percent	0-6	Owning
TBC456	Re-alert Time Interval <i>Time interval between audible alerts when the re-alerting functionality is enabled. See also CFG51.</i>	seconds	10-600	Operating
TBC479	Restricted Speed Canada <i>Maximum speed allowed under Restricted Speed rules, used on track where the country of operation is Canada</i>	mph	1-20	Operating
		km/h	1-32	

All CFG parameters listed in the table below are boolean.

Table 3.2: Railroad Functional Parameters

CFG#	Description	Railroad
CFG1	Allow Movement in the Opposite Direction from that Authorized by a Directional Movement Authority	Operating railroad issuing the movement authority
CFG2	Only Enforce Restricted Speed in Yard Limits that Require Form-Based and Signal Authority <i>True: Generate restricted speed target based on indication of signal controlling entry to Yard Limits only on track that is signaled and requires form-based authority in the train's direction of movement (i.e. ABS or equivalent)</i> <i>False: Generate restricted speed target based on indication of signal controlling entry to Yard Limits on track that is signaled in the train's direction of movement regardless of whether form-based authority is required (i.e. CTC or equivalent and ABS or equivalent)</i> <i>See also CFG10</i>	Operating railroad of the signal controlling movement in the yard limits
CFG3	Allow Movement in a Work Zone when the Train Orientation Changes	Operating railroad issuing the work zone bulletin
CFG5	Prompt Crew For Removal of Previously Acknowledged Bulletin	Operating railroad issuing the bulletin
CFG6	Horn Protection Sequenced Activation <i>True: The on-board segment activates the horn in the sequence specified in the I-ETMS Protect On-Board Segment Requirement Specification</i> <i>False: The on-board segment activates the horn in a continuous blast</i>	Operating railroad of the grade crossing

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CFG#	Description	Railroad
CFG7	Energy Management Supported	Owning
CFG8	Send Messages to CI BOS <i>Identifies if the railroad operates with a crew interaction back office server.</i>	Employing, Owning, Operating
CFG9	Send PTC Interaction (02082) Message	Operating
CFG10	Enforce Restricted Speed in Yard Limits for Signal Indication in SG-4 <i>True: Generate restricted speed target when signal controlling entry to Yard Limits has an indication of dark, unknown, SG-1, SG-2, SG-3, or SG-4</i> <i>False: Generate restricted speed target when signal controlling entry to Yard Limits has an indication of dark, unknown, SG-1, SG-2, or SG-3</i> <i>See also CFG2</i>	Operating railroad of the signal controlling movement in the yard limits
CFG12	Send EMT/PSS Authority Used Notification when entering main track or passing signal at stop on authority manually indicated by train crew	Operating railroad of the track being entered
CFG13	Enable Enter Main Track Prompting	Operating railroad of the track being entered
CFG14	Prevent Crew From Modifying Summary Consist Data	Operating
CFG15	Support 802.1q VLAN	Owning
CFG16	Make the target generated for delayed in block on track that is signaled and requires form based authority in the train's direction of movement a restricted speed target <i>True: When the on-board segment's delayed in block functionality is invoked on track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the delayed in block target generated is a restricted speed target</i> <i>False: When the on-board segment's delayed in block functionality is invoked on track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), the delayed in block target generated is not a restricted speed target</i> <i>See also TBC283</i>	Operating railroad of the signal controlling movement in the block
CFG17	Enable prompting to remove delayed in block target when the next facing signal in advance of the train has a status in group SG-3	Operating railroad of the signal controlling movement in the block
CFG18	Loaded car count includes isolated locomotives and loaded cars <i>True: Locomotive consist sent in a 01030 message or modified by crew must include any isolated locomotive in the loaded car count and as a locomotive indicating isolated</i> <i>False: Locomotive consist sent in a 01030 message or modified by crew must include any isolated locomotive only as a locomotive.</i> <i>Note: Any dead-in-tow locomotives should be treated the same as isolated locomotives. However, if railroad operations require dead-in-tow locomotives to be treated as loaded cars, the braking algorithm will maintain safe braking curves.</i>	Operating railroad that provided the summary consist data
CFG19	Trailing tonnage includes isolated locomotives and all cars <i>True: Consist sent in a 01030 message or modified by crew must include the weight of any isolated locomotive with the trailing tonnage</i> <i>False: Consist sent in a 01030 message or modified by crew must not include the weight of any isolated locomotive with the trailing tonnage.</i>	Operating railroad that provided the summary consist data

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CFG#	Description	Railroad
CFG20	Axle count includes all locomotives and all cars	Operating railroad that provided the summary consist data
CFG21	Generate a pass signal at stop target beyond a signal displaying stop on track that is signaled and requires form based authority in the train's direction of movement <i>True: On track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), allow the train to move past the signal at stop no more than TBC284 feet if the crew indicates that they are unable to contact the dispatcher and can see the signal at stop. Prompt the crew for authority to proceed to the next signal if the train is stopped within TBC285 feet for TBC286 seconds.</i> <i>False: On track that is signaled and requires form based authority in the train's direction of movement (i.e. ABS or equivalent), use the same pass signal at stop functionality the on-board segment performs on track that is signaled and does not require form based authority in the train's direction of movement (i.e. CTC or equivalent) See also TBC284, TBC285, and TBC286</i>	Operating railroad of the signal
CFG22	Enforce Work Zone for Head-End Only <i>Enabled: The on-board segment generates the initial zero speed target over the work zone limits as a head-end only target</i> <i>Disabled: The on-board segment generates the initial zero speed target over the work zone limits as an entire train target</i> <i>Note that the setting of this parameter does not affect enforcement of any speed restriction over the work zone limits</i>	Operating railroad issuing the work zone bulletin
CFG23	Prompt for Ability to Clear Work Zone or Permission to Proceed Through Work Zone Prior to Effective Time <i>Enabled: When the on-board segment prompts for a work zone prior to the effective time of that work zone, the prompt will ask the crew to acknowledge that they can either clear the limits of the work zone before it becomes effective or that they have permission to proceed through the work zone.</i> <i>Disabled: There is no difference between the work zone prompts displayed prior to the effective time of the work zone and those displayed when the work zone is in effect.</i> <i>See also TBC290</i>	Operating railroad issuing the work zone bulletin
CFG24	Allow MDM Client logging of File Data Request (02103) and File Data (01103) messages	Owning
CFG25	Allow MDM Client logging of File Data Request (01122) and File Data (02122) messages	Owning
CFG27	Determine enforcement of grade crossing malfunction bulletin by required speed at crossing <i>True: For a grade crossing on track with 1) a country of operation other than Canada, or 2) country of operation Canada and CFG61 disabled, the on-board segment prompts for the required speed at the crossing and adjusts its targeting and enforcement accordingly</i> <i>False: For a grade crossing on track with 1) a country of operation other than Canada, or 2) country of operation Canada and CFG61 disabled, the on-board segment does not prompt for the required speed at the crossing. The on-board segment enforces a stop prior to the crossing.</i>	Operating
CFG28	Generate a stop target at the next facing signal in the direction of a reverse movement <i>True: The on-board segment generates a stop target at the next facing signal in the direction of a reverse move on signaled track</i>	Operating

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CFG#	Description	Railroad
	that does not require form based authority. <i>False: The on-board segment does not generate a stop target at the next facing signal in the direction of a reverse move on signaled track that does not require form based authority</i>	
CFG29	Retain Signal Target Status <i>Enabled: The on-board segment retains signal target status upon transition to CUT OUT or RESTRICTED state and re-instates the signal target upon re-establishing a known locomotive track location, assuming all conditions for doing so are met as detailed in the on-board segment requirements.</i> <i>Disabled: The on-board segment does not retain signal target status upon transition to CUT OUT or RESTRICTED state.</i> <i>An operating railroad may choose to disable this functionality, for example, to ensure consistent behavior of the on-board segment on locomotives equipped with active and passive wheel tachometers. An active tachometer is a condition for retaining signal target status.</i>	Operating
CFG30	Enable Crew Selection from Multiple Pending Prompts with "View All Prompts" Key	Operating
CFG33	Enforce restricted speed for longer of train length or TBC296 ft after cab signal upgrades <i>Enabled: The on-board segment enforces restricted speed for the longer of train length or TBC296 ft after the cab signal upgrades from restricting.</i> <i>Disabled: The on-board segment immediately enforces the speed associated with the new cab signal status when the cab signal upgrades from restricting.</i>	Operating
CFG34	Enable MTEA Speed Enforcement for Owning Railroad <i>The on-board segment enforces MTEA speed when the on-board segment is Disengaged and not navigating when CFG34 is enabled by the owning railroad and CFG35 is enabled by all operating railroads selected for initialization. Otherwise, the on-board segment does not enforce MTEA speed.</i>	Owning
CFG35	Enable MTEA Speed Enforcement for Operating Railroad <i>The on-board segment enforces MTEA speed when the on-board segment is Disengaged and not navigating when CFG34 is enabled by the owning railroad and CFG35 is enabled by all operating railroads selected for initialization. Otherwise, the on-board segment does not enforce MTEA speed.</i>	Operating
CFG36	Generate shoving move speed restriction <i>Enabled: The on-board segment generates a shoving move speed restriction target when the train orientation is "reverse".</i> <i>Disabled: The on-board segment does not generate a shoving move speed restriction target when the train orientation is "reverse".</i> <i>See also TBC417 and TBC418.</i>	Operating
CFG37	Enable IC3 Functionality <i>Enabled: The on-board segment evaluates the IC3 CRC and IC3 Uniqueness Index fields in the messages it receives from the back office. The on-board segment sends a Poll Response (02021) message in response to the Office Segment Poll (01021) message.</i> <i>Disabled: The on-board segment ignores the IC3 CRC and IC3 Uniqueness Index fields in the messages it receives from the back office. The on-board segment does not send a Poll Response (02021) message in response to the Office Segment Poll (01021) message.</i>	Operating

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CFG#	Description	Railroad
CFG38	Target Dark Signal in Advance of SG-5 Signal Immediately <i>Enabled: When a dark signal follows a signal with an indication in SG-5 on the train's calculated route, the on-board segment will generate a target for the dark signal immediately without waiting for the leading edge of the train to pass the last facing signal in approach to the dark signal.</i> <i>Disabled: When a dark signal follows a signal with an indication in SG-5 on the train's calculated route, the on-board segment will wait until TBC269 seconds have elapsed since the leading edge of the train passed the last facing signal in approach to the dark signal before generating a target for the dark signal.</i> CFG38 is provided to accommodate railroads which may have mapped a signal indication which would normally be in SG-3 to SG-5. Enabling CFG38 ensures that the on-board segment targets a dark signal following such a signal early enough to provide timely enforcement at the dark signal.	Operating
CFG40	Restrict Navigation Module Messaging to NMEA <i>Enabled: The on-board segment restricts messaging with the navigation modules to use NMEA formatted messages only.</i> <i>Disabled: The on-board segment allows additional messaging formats such as S-9103 with the navigation modules.</i>	Owning
CFG41	Disallow Acceptance of Wayside Signal Status behind the Calculated Train Leading Edge Location When Stopped <i>Enabled: When the locomotive has been stopped for at least TBC412 seconds, the on-board segment will accept WIUStatus messages from signals on the train's calculated route between the leading edge of the train and the end of the train's calculated route.</i> <i>Disabled: When the locomotive has been stopped for at least TBC412 seconds, the on-board segment will accept WIUStatus messages from signals on the train's calculated route between a point at the calculated position uncertainty behind the leading edge of the train and the end of the train's calculated route.</i>	Operating
CFG42	Enable generation of end of form based authority on signaled track target <i>Enabled: The on-board segment generates a zero (0) mph 'end of form based authority on signaled track' target at the end of an authority on track that is signaled in the train's direction of movement and does not require form based authority in the train's direction of movement and no facing signal or additional form based authority is coincident with the end of that authority to provide authority for the train to exit that authority.</i> <i>Disabled: The on-board segment does not generate a zero (0) mph 'end of form based authority on signaled track' target. A railroad may choose to disable this CFG, for example, if their operating rules allow a train to proceed based on the indication of the last facing signal within the limits of an authority on track that is signaled in the train's direction of movement and does not require form based authority in the train's direction of movement even if there is no facing signal coincident with the exit limit of that authority.</i>	Operating
CFG43	Limit generation of end of form based authority on signaled track targets to bidirectional authorities <i>Enabled: When CFG42 and CFG43 are enabled, the on-board segment generates a zero (0) mph 'end of form based authority on signaled track' target only at the end of a bidirectional authority.</i>	Operating

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CFG#	Description	Railroad
	<i>Disabled: When CFG42 is enabled and CFG43 is disabled, the board segment generates a zero (0) mph 'end of form based authority on signaled track' target at the end of directional authorities and bidirectional authorities.</i>	
CFG44	Enable logging of messages exchanged between PTC and EM <i>Enabled: Messages exchanged between the on-board segment and Energy Management are logged.</i> <i>Disabled: Messages exchanged between the on-board segment and Energy Management are not logged.</i>	Owning
CFG45	Allow the Date to be displayed in the upper left corner of the display	Owning
CFG46	Automatic Parking Function	Owning
CFG48	Operating railroad elects to receive Onboard Train Consist (02032) message <i>Enabled: The on-board segment sends the Onboard Train Consist (02032) message to each operating railroad for which CFG48 is enabled.</i> <i>Disabled: The on-board segment does not send the Onboard Train Consist (02032) to an operating railroad if CFG48 is disabled for that railroad and that railroad is not designated as providing summary consist information to the on-board segment.</i>	Operating
CFG49	When enabled, the rear end location in the Locomotive Position Report (02080) message will be set to the point at the calculated position uncertainty behind the trailing edge of the train	Operating
CFG50	Include valid LIG BPP2 as input to Front Brake Pipe Pressure determination <i>Enabled: On a locomotive with a LIG control system and available BPP2 input, the on-board segment includes valid BPP2 input in the determination of a Front Brake Pipe Pressure value.</i> <i>Disabled: On a locomotive with a LIG control system and available BPP2 input, the on-board segment omits the BPP2 input from the determination of a Front Brake Pipe Pressure value.</i>	Owning
CFG51	Re-alert Enabled <i>Enabled: The on-board segment sounds additional audible alerts after the initial audible alert when the locomotive is stopped with the reverser handle in the forward or reverse position and a prompt for a zero (0) mph target is displayed.</i> <i>Disabled: The on-board segment does not sound additional audible alerts after the initial audible alert when a prompt for a zero (0) mph target is displayed.</i> <i>See also TBC456.</i>	Operating
CFG52	Perform multi-parameter consist checks <i>Enabled: The on-board segment checks combinations of consist values to ensure they are consistent and prompts the crew to correct errors when they are detected.</i> <i>Disabled: The on-board segment does not perform the consist checks described above.</i>	Operating railroad that provided the summary consist data
CFG56	Licensed Feature Civil Speed Enabled	Owning
CFG61	Crossing Protection Canada <i>Enabled: For a grade crossing on track with country of operation Canada and CFG61 enabled, the on-board segment prompts for the required protection at the crossing, either manual protection required or maximum authorized speed.</i> <i>Disabled: For a grade crossing on track with country of operation Canada and CFG61 disabled, the on-board segment either prompts for the required speed at the crossing if CFG27 is enabled (either stop and warn required, TBC30 mph or maximum authorized speed) or does not prompt for required speed at the crossing if CFG27 is</i>	Operating

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CFG#	Description	Railroad
	<i>disabled.</i>	
CFG65	Reverse Move Restricted Speed Canada <i>Enabled: When the country of operation is Canada and CFG65 is enabled, the on-board segment will enforce restricted speed when the train orientation changes in a signal block and when the train has moved more than TBC310 feet opposite the direction in which the train entered the signal block as it does when the country of operation is other than Canada.</i> <i>Disabled: When the country of operation is Canada and CFG65 is disabled, the on-board segment will not enforce restricted speed when the train orientation changes in a signal block nor when the train has moved more than TBC310 feet opposite the direction in which the train entered the signal block</i>	Operating
CFG66	PSS No Stop Required <i>Enabled: When the country of operation is Canada and CFG66 is enabled for the operating railroad, the on-board segment does not require the train to be stopped before applying a movement authority dataset of type Pass Signal at Stop, CROR PSS No Equipment or CROR PSS Equipment.</i> <i>Disabled: When the country of operation is Canada and CFG66 is disabled for the operating railroad, the on-board segment requires the train to be stopped before applying a movement authority dataset of type Pass Signal at Stop, CROR PSS No Equipment or CROR PSS Equipment.</i>	Operating
CFG67	PSS Manual Prompt Speed Selection <i>Enabled: When the country of operation is Canada and CFG67 is enabled for the operating railroad, the on-board segment will prompt the crew to indicate the speed to be enforced after authorized to pass a signal at stop.</i> <i>Disabled: When the country of operation is Canada and CFG67 is disabled for the operating railroad, the on-board segment will not prompt the crew to indicate the speed to be enforced after authorized to pass a signal at stop. The prompts displayed for passing a signal at stop match those displayed when the country of operation is other than Canada.</i>	Operating
CFG68	Upgrade Signal in Bidirectional Authority <i>Enabled: The on-board segment replaces a zero (0) mph signal target with a restricted speed signal target or a zero (0) mph unknown signal target with a restricted speed unknown signal target at an absolute non-interlocking signal when</i> • <i>the leading edge of the train is within TBC40 mph feet plus the calculated position uncertainty of a signal and</i> • <i>that signal is on track with country of operation Canada and</i> • <i>that signal is on track that is signaled in the train's direction of movement but does not require form-based authority in the train's direction of movement and</i> • <i>that signal is inside but not coterminous with the limits of a bidirectional authority and</i> • <i>that signal is not the last signal on the train's calculated route in approach to:</i> ○ <i>the end of the bidirectional authority or</i> ○ <i>the end of track signaled in the train's direction of movement</i> ○ <i>the end of PTC track or</i> ○ <i>the end of the train's calculated route.</i> <i>Disabled: The on-board segment requires authority to pass signal at stop for a signal with the criteria listed above in order to replace a</i>	Operating

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CFG#	Description	Railroad
	<i>zero (0) mph signal target or a zero (0) mph unknown signal target at that signal.</i>	
CFG69	Current speed display with one decimal resolution. <i>Enabled: The onboard current speed is displayed rounded down to one decimal.</i> <i>Disabled: The onboard current speed is displayed rounded down to a whole number.</i>	Employing, Owning
CFG70	Acceleration rate display. <i>Enabled: The onboard acceleration rate is displayed on top the CDU, below the current speed.</i> <i>Disabled: The onboard will not display the acceleration rate.</i>	Employing, Owning

3.2 Train Handling Exceptions

Configuration File: `params_THE_RR.cfg`

NOTE The **RR** characters in the actual configuration files are replaced with the individual railroad SCAC code.

Each version of the `params_THE_RR.cfg` file contains railroad specific Train Handling Exception parameters that are configured by the individual ITC member railroads. Train Handling Exceptions always use the configuration of the employing railroad. This includes defining the bounds of the exceptions and turning on and off notifications.

Note that for railroads operating in metric, all parameters representing speeds must be entered as kilometers per hour within the ranges defined in the table below for applicable parameters.

Table 3.3: Train Handling Exception TBC Values

TBC#	Description	Units	Valid Range (min-max)
TBC4	Bail-off of Independent Brakes Maximum BPP	percent	0-100
TBC5	Bail-off of Independent Brakes Minimum BCP	percent	0-100
TBC6	Bail-off of Independent Brakes Minimum Speed	mph	0-125
		km/h	0-201
TBC7	Bail-off of Independent Brakes Timeout	seconds	1-1000
TBC8	Brake Pipe Continuity Determination Time Interval	seconds	1-1000
TBC9	Brake Pipe Continuity Minimum Pressure Threshold	percent	0-100
TBC10	Brake Pipe Continuity Pressure Differential	percent	0-100
TBC13	Cab Signal Exception Handling Speed	mph	0-125
		km/h	0-201
TBC43	Dynamic Brake Exception Determination Time	seconds	1-1000
TBC49	Crew Induced Emergency Braking Maximum Speed	mph	0-125
		km/h	0-201
TBC50	Crew Induced Emergency Braking Minimum Speed	mph	0-125
		km/h	0-201
TBC52	Excessive Tractive Effort Maximum Speed	mph	0-125
		km/h	0-201
TBC53	Excessive Tractive Effort Minimum Throttle Position	n/a	0-8
TBC63	Heavy Reduction Braking Timeout	seconds	0-1000
TBC64	Heavy Reduction Maximum Pressure Threshold	percent	0-100
TBC65	Heavy Reduction Minimum Speed	mph	0-125
		km/h	0-201
TBC68	Independent Braking Maximum Time Limit	seconds	1-1000
TBC69	Independent Braking Minimum Pressure Threshold	percent	0-100

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TBC#	Description	Units	Valid Range (min-max)
TBC70	Independent Braking Minimum Speed	mph	0-125
		km/h	0-201
TBC71	Independent Braking Minimum Time Limit	seconds	0-1000
TBC72	Independent Braking on Heavy Grade Minimum Percent	percent	0-100
TBC73	Independent Braking on Heavy Grade Minimum Pressure	percent	0-100
TBC74	Independent Braking on Heavy Grade Minimum Speed	mph	0-125
		km/h	0-201
TBC75	Independent Braking on Heavy Grade Timeout	seconds	1-1000
TBC76	Independent Braking Timeout	seconds	1-1000
TBC79	Long Duration Braking Cycle	n/a	1-2
TBC80	Long Duration Braking Cycle Time Limit	seconds	1-240
TBC81	Long Duration Braking Pressure Drop	psi	1-6
TBC82	Long Duration Braking Pressure Increase	psi	1-6
TBC117	Multiple Reductions Braking Determination Time Interval	seconds	1-120
TBC118	Multiple Reductions Braking Pressure Threshold	psi	1-18
TBC125	Power Braking Maximum Pressure Threshold	percent	0-100
TBC126	Power Braking Minimum Speed	mph	0-125
		km/h	0-201
TBC127	Power Braking Minimum Throttle Position	n/a	0-8
TBC128	Power Braking Timeout	seconds	1-1000
TBC142	Short Duration Braking Cycle	n/a	1-3
TBC143	Short Duration Braking Cycle Time Limit	seconds	1-180
TBC144	Short Duration Braking Pressure Drop	psi	1-6
TBC145	Short Duration Braking Pressure Increase	psi	1-6
TBC149	Standstill with DC Power Maximum Grade	percent	0-100
TBC150	Standstill with DC Power Minimum Notch	n/a	0-8
TBC151	Standstill with DC Power Minimum Pressure	percent	0-100
TBC152	Standstill with DC Power Speed	mph	0-125
		km/h	0-201
TBC153	Stopped with Brakes Released Minimum Pressure	percent	0-100
TBC154	Stopped with Brakes Released Speed	mph	0-125
		km/h	0-201
TBC155	Stopped with Brakes Released Timeout	seconds	1-1000
TBC164	Throttle Modulation Minimum Notch Change	n/a	1-8
TBC165	Throttle Modulation Notch Change Time Interval	seconds	1-1000
TBC169	Train Induced Emergency Braking Maximum Speed	mph	0-125
		km/h	0-201
TBC170	Train Induced Emergency Braking Minimum Speed	mph	0-125
		km/h	0-201

All THE parameters listed in the table below are boolean.

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- When a THE##D parameter is set to true, detection of that train handling exception is enabled.
- When a THE##P parameter is set to true, prompting of the train crew for that train handling exception is enabled.
- When a THE##N parameter is set to true, office notification for that train handling exception is enabled.

Table 3.4: Train Handling Exception (THE) Values

THE#	Description
THE01D	Power Braking – Detect
THE01P	Power Braking – Prompt
THE01N	Power Braking – Notify
THE02D	Heavy Reduction Braking – Detect
THE02P	Heavy Reduction Braking – Prompt
THE02N	Heavy Reduction Braking – Notify
THE03D	Brake Pipe Continuity – Detect
THE03P	Brake Pipe Continuity – Prompt
THE03N	Brake Pipe Continuity – Notify
THE04D	Independent Braking (Type 1) – Detect
THE04P	Independent Braking (Type 1) – Prompt
THE04N	Independent Braking (Type 1) – Notify
THE05D	Independent Braking (Type 2) – Detect
THE05P	Independent Braking (Type 2) – Prompt
THE05N	Independent Braking (Type 2) – Notify
THE06D	Independent Braking on Heavy Grade – Detect
THE06P	Independent Braking on Heavy Grade – Prompt
THE06N	Independent Braking on Heavy Grade – Notify
THE07D	Failure to Bail-off Independent Brakes after Automatic Brake Application – Detect
THE07P	Failure to Bail-off Independent Brakes after Automatic Brake Application – Prompt
THE07N	Failure to Bail-off Independent Brakes after Automatic Brake Application – Notify
THE08D	Crew Induced Emergency Braking – Detect
THE08P	Crew Induced Emergency Braking – Prompt
THE08N	Crew Induced Emergency Braking – Notify
THE09D	Dynamic Brake (Type 1) – Detect
THE09P	Dynamic Brake (Type 1) – Prompt
THE09N	Dynamic Brake (Type 1) – Notify
THE10D	Dynamic Brake (Type 2) – Detect
THE10P	Dynamic Brake (Type 2) – Prompt
THE10N	Dynamic Brake (Type 2) – Notify
THE11D	Cab Signal – Detect
THE11P	Cab Signal – Prompt
THE11N	Cab Signal – Notify
THE12D	EOT Failure – Detect
THE12P	EOT Failure – Prompt
THE12N	EOT Failure – Notify
THE13D	Use of Excessive Tractive Effort when Starting Train – Detect
THE13P	Use of Excessive Tractive Effort when Starting Train – Prompt
THE13N	Use of Excessive Tractive Effort when Starting Train – Notify
THE14D	Stopped with Brakes Released – Detect
THE14P	Stopped with Brakes Released – Prompt
THE14N	Stopped with Brakes Released – Notify
THE15D	DC Locomotive Holding Train at Standstill with Power and Automatic Brakes Released – Detect

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THE#	Description
THE15P	DC Locomotive Holding Train at Standstill with Power and Automatic Brakes Released – Prompt
THE15N	DC Locomotive Holding Train at Standstill with Power and Automatic Brakes Released – Notify
THE16D	Throttle Modulation – Detect
THE16P	Throttle Modulation – Prompt
THE16N	Throttle Modulation – Notify
THE17D	Train Induced Emergency Braking – Detect
THE17P	Train Induced Emergency Braking – Prompt
THE17N	Train Induced Emergency Braking – Notify
THE18D	Short Duration Braking Cycle – Detect
THE18P	Short Duration Braking Cycle – Prompt
THE18N	Short Duration Braking Cycle – Notify
THE19D	Long Duration Braking Cycle – Detect
THE19P	Long Duration Braking Cycle – Prompt
THE19N	Long Duration Braking Cycle – Notify
THE20D	Multiple Brake Reductions – Detect
THE20P	Multiple Brake Reductions – Prompt
THE20N	Multiple Brake Reductions – Notify

3.3 BOS Polygons

Configuration File: BOSPolygon_RR.cfg

NOTE The **RR** characters in the actual configuration files are replaced with the individual railroad SCAC code.

Each version of the BOSPolygon_RR.cfg file contains railroad specific geo-spatial polygons that are used to determine when a locomotive is within the railroad's operating range.

Based on the PTC data model, railroads define one or more geo-spatial polygons that may have one or more BOS EMP Addresses per polygon. This allows a railroad to have BOS instances set up to serve a specific geographic area.

NOTE A maximum of 100 polygons can be defined in a file.

Table 3.5: BOSPolygon_RR.cfg Parameters

Parameter	Description
rr_name	RR SCAC name without spaces
default_addr	Default BOS EMP address <i>If the railroad is configured to have a CI BOS, this default address must be the address of the CI BOS</i>
mdm_def_addr	Default MDM EMP address
version	Maximum 4 character alphanumeric
Up to 100 polygons in either of the two formats below:	
poly_lower_left_corner	Comma separated (no spaces) coordinates, each can be maximum of 8 byte float
poly_upper_right_corner	Comma separated (no spaces) coordinates, each can be maximum of 8 byte float
poly_addr_list	Maximum 4 comma separated (no spaces) address in a polygon <i>If the railroad is configured to have a CI BOS, this list must contain one address which is the address of the CI BOS</i>

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Parameter	Description
Or	
[polygon3]	
poly_vertices	max 15 min 3 vertices separated by 'v' for each vertex comma separated (no spaces) coordinates each can be maximum of 8 byte float
poly_addr_list	Maximum 4 comma separated (no spaces) address in a polygon <i>If the railroad is configured to have a CI BOS, this list must contain one address which is the address of the CI BOS</i>

3.4 Communication Parameters

Configuration File: `params_comm_RR.cfg`

NOTE The **RR** characters in the actual configuration files are replaced with the individual railroad SCAC code.

Each version of the `params_comm_RR.cfg` file contains railroad specific communication parameters. These parameters include the IP address and port needed to connect to the Application Gateway, Energy Management Client, and Event Recorder. These parameters are defined in AAR Manual of Standards and Recommended Practices, Class D Messaging.

Table 3.6: params_comm_RR.cfg Parameters

Parameter	Description	Units	Valid Range
local_ip	In Server mode, connection Request is listened on this port	n/a	
dest_ip	In client mode, connection attempt is made for this IP	n/a	
server_mode	Y=Server Mode N=Client Mode	n/a	Y or N
link_mode	BIDIRECT = Bidirectional RCV = Receive Only SEND = Send Only	n/a	BIDIRECT RCV SEND
data_ack_enable	Y = Data ACK expected N = Data ACK not expected	n/a	Y or N
logging_on	Y = Class D msg logged at SYS level N = Class D msg logged at MDATA level	n/a	Y or N
k_a_timer	Keep Alive Interval 0 = KA disabled	ms	0 - 60000
data_ack_timeout	Data ACK timeout	ms	0 - 60000
ka_ack_timeout	Keep Alive ACK timeout	ms	0 - 60000
nak_retries	NAK retries limit for Data Msg	n/a	0 - 10
cnt_timeout	Connection Attempt Timeout	ms	1 - 60000
cnt_delay	Connection Delay	ms	1 - 60000
cnt_retry	Connection Retry limit, resets at successful connection	n/a	-1 to 10000
recnt_retry	Re-Connect Retry limit	n/a	-1 to 10000
persistence_on	Y = Persistence ON N = Persistence OFF	n/a	Y or N
retransmit_delay	Retransmit delay after getting a NAK	ms	0 - 10000

3.5 VLAN Parameters

Configuration File: `params_vlan_RR.cfg`

NOTE The **RR** characters in the actual configuration files are replaced with the individual railroad SCAC code.

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Each version of the `params_vlan_RR.cfg` file contains railroad specific VLAN configuration parameters to configure the ethernet switch in the RSM to use 802.11q VLANs.

The format of the `params_vlan_RR.cfg` file is shown below:

```
[port default]

# for each port
<port ID>, <vlan number>

[vlan table]
#for each vlan
<vlan number>untag<port ID list>
<vlan number>forward<port ID list>
```

The first part of the file lists the default VLAN number for each port in I-ETMS on-board segment. The current list of external RSM port IDs are J1, J2, J3, J4, J5 and J5b. The current list of internal card slot (left to right) port IDs are SL1, SL2, SL3, SL4, SL7, SL8, SL9 and SL10. Up to six VLANs may be defined with VLAN numbers in the range 1-254. VLAN 101 is reserved for PTC traffic. The railroad can configure up to five additional VLANs.

The second part of the file defines the tagging and forwarding functions for each VLAN. The port ID list is a comma separated list of port IDs for which the “untag” or “forward” parameter is set to true. If the parameter is false for all port IDs, then the port ID list is empty.

The IEEE 802.11q standard for VLANs defines a “tag” added to the beginning of a packet that defines to which VLAN that packet belongs. Packets can be passed to the Ethernet switch tagged with their VLAN, or if the packet does not have tag the switch will tag it with the default tag for that port.

If the “untag” parameter is set to true for a port, then packets belonging to that VLAN that are sent out that port have the tag stripped off. If the “untag” parameter is set to false, then packets belonging to that VLAN will be sent out that port with tags. The I-ETMS on-board segment application is not expecting tags, so the “untag” parameter should be set to true for all I-ETMS ports on the PTC VLAN.

If the “forward” parameter is set to true for a port, then packets belonging to that VLAN are sent out that port. If the “forward” parameter is set to false for a port, then packets for that VLAN are not sent out that port.

3.6 Train Restriction Types

Configuration File: `train_restriction_types_RR.cfg`

NOTE The `RR` characters in the actual configuration files are replaced with the individual railroad SCAC code.

The `train_restriction_types_RR.cfg` file contains textual descriptions for the train restriction types that are specific to an individual railroad. The numeric value assigned to each train restriction type corresponds to the value assigned in the summary consist messages exchanged with the back office and to the value assigned in the track data for the train restriction type.

The format of the `train_restriction_types_RR.cfg` file is shown below:

```
[TRAIN_RESTRICTION_TYPES]

#for each train restriction type
<train restriction type number> <train restriction type description>
```

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Any train_restriction_types_RR.cfg file that defines more than one description for a train restriction type number is considered invalid.

Any train_restriction_types_RR.cfg file that contains a train restriction type description longer than 30 characters is considered invalid.

Any train_restriction_types_RR.cfg file that defines a train restriction type description for a type number outside the range of 17-32 is considered invalid.

Any train_restriction_types_RR.cfg file that defines a train restriction type without a corresponding description is considered invalid.

3.7 License

Configuration File: license_RR.cfg

NOTE The RR characters in the actual configuration files are replaced with the individual railroad SCAC code.

Each version of the license_RR.cfg file contains license codes for I-ETMS on-board segment functionality for which the railroad has purchased a license. Below the license codes, any TBC and CFG values will be listed for the applicable licenses.

The format of the license_RR.cfg file is shown below:

```
[license table]

# for each license
<license string>

[configurable parameter table]

#configurable items
<TBC number and name><value>
<CFG number and name><value>
```

The license string is a variable length string representation of a hash provided by Wabtec to the licensee. For more information on Licensed Feature configurable items, see the applicable addendum specification. Only configurable items that differ from the defaults defined in the applicable addendum specification are needed to be included.

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/// 4 Onboard EMP Messaging Names

All messages exchanged between the I-ETMS TMC and off-board entities are expected to conform to the AAR Edge Message Protocol standard. Table 4.1: Onboard EMP Messaging Names lists the messaging names to be used for the components hosted on the I-ETMS TMC.

Note that not all of these components are considered part of the I-ETMS on-board segment. They are defined here to avoid conflicts between the messaging names of the I-ETMS components and the other hosted components.

Note that the values in Table 4.1: Onboard EMP Messaging Names are not stored in a configuration file and may only be changed with an I-ETMS on-board software release.

Table 4.1: Onboard EMP Messaging Names

Onboard Component		Messaging Name
Application Gateway		ag
Application Gateway Business Application Manager		ag.mdm
Crash Hardened Memory		chm
Energy Management Business Application Manager		em.mdm
Onboard Energy Management		em
Onboard ITC	For communication with MDM Server	itc.mdm <i>OR</i> mdm
	For all other communication	itc

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/// 5 Configuration File Formatting

Wabtec formats the configuration files it releases with Unix style line endings, i.e. line feed only. This convention should be observed in any subsequent editing of configuration files in order to maintain interoperability.

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